



# ING. MECATRÓNICA

**Trabajo de Integración Curricular previo a la obtención del título de  
Mechatronics Engineer.**

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Desarrollo de un sistema de visión artificial para la obtención estandarizada de imágenes ecográficas hepáticas de relevancia clínica

Development of an Artificial Vision System for the Standardized Acquisition of Clinically Relevant Hepatic Ultrasound Images

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### **Aprobación del Tutor del Trabajo de Integración Curricular**

Yo Lorena de los Ángeles Guachi Guachi, en calidad de tutor del Trabajo de Integración Curricular denominado: *Desarrollo de un sistema de visión artificial para la obtención estandarizada de imágenes ecográficas hepáticas de relevancia clínica*, realizado por Julian Andre Guapaz Pozo, ha sido orientado y revisado en todas sus partes y considero que cumple los requisitos establecidos por la institución. En virtud de ello doy mi aprobación a fin de continuar con el proceso académico correspondiente.

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## Resumen

Las imágenes ecográficas se utilizan ampliamente por su bajo costo, portabilidad y carácter no invasivo; sin embargo, su adquisición depende en gran medida de la experiencia del operador. Este trabajo presenta un sistema de visión artificial para asistir a usuarios con experiencia limitada en la adquisición de cuatro vistas ecográficas hepáticas: *longitudinal*, *transversal*, *oblicua* y *hepatorrenal*. El sistema emplea dos estrategias para identificar *frames* informativos. Para la *vista longitudinal*, DeepLabV3+ segmenta la región ecográfica de interés (ROI) y el hígado, mientras que U-Net segmenta el lumen anecoico (LA); a partir de estas regiones se extraen características morfológicas, estadísticas y de textura para determinar la informatividad del *frame*. Para las vistas *transversal*, *oblicua* y *hepatorrenal*, DINOv2-Small extrae directamente de cada *frame* representaciones visuales profundas, posteriormente clasificadas mediante Random Forest, regresión logística y k-NN, respectivamente.

El sistema se desarrolló con 4,040 *frames* adquiridos de tres voluntarios y se evaluó de forma independiente sobre 483 resultados de análisis asociados a *frames* de 12 secuencias ecográficas, correspondientes a tres adquisiciones de cada vista realizadas por un voluntario sin experiencia previa en ecografía tras aproximadamente 15 min de entrenamiento. El *ground truth* fue establecido por un radiólogo experto, quien clasificó cada *frame* como informativo o no informativo según la visualización de las estructuras anatómicas requeridas para cada vista. El análisis binario incluyó 322 resultados; los 161 restantes se conservaron para el análisis de incertidumbre. En el conjunto binario, el sistema identificó correctamente el 37,27% (60/161) de los *frames* informativos y el 90,68% (146/161) de los no informativos, correspondientes a una sensibilidad y especificidad globales de 37,27% y 90,68%, respectivamente. El desempeño por vista se evaluó además mediante *precision*, *recall*, F1-score y *balanced accuracy*. Los F1-scores fueron 0,333, 0,564, 1,000 y 1,000, y los valores de *balanced accuracy* fueron 0,500, 0,625, 1,000 y 1,000 para las vistas *longitudinal*, *transversal*, *oblicua* y *hepatorrenal*, respectivamente. Estos resultados preliminares delimitan la viabilidad técnica del enfoque en el conjunto evaluado y justifican su validación futura con más pacientes, operadores y entornos de adquisición.

**Palabras clave:** ecografía hepática, visión artificial, identificación de imágenes informativas.

### Abstract

Ultrasound images are widely used because of their low cost, portability, and non-invasive nature; however, their acquisition depends heavily on the operator's experience. This work presents a computer vision system to assist users with limited experience in acquiring four hepatic ultrasound views: *longitudinal*, *transverse*, *oblique*, and *hepatorenal*. The system uses two strategies to identify informative *frames*. For the *longitudinal view*, DeepLabV3+ segments the ultrasound region of interest (ROI) and the liver, while U-Net segments the anechoic lumen (LA); morphological, statistical, and texture features are then extracted from these regions to determine *frame* informativity. For the *transverse*, *oblique*, and *hepatorenal views*, DINOv2-Small directly extracts deep visual representations from each *frame*, which are subsequently classified using Random Forest, logistic regression, and k-NN, respectively.

The system was developed using 4,040 *frames* acquired from three volunteers and independently evaluated on 483 analysis outputs associated with *frames* from 12 ultrasound sequences, comprising three acquisitions of each view performed by a volunteer with no previous ultrasound experience after approximately 15 min of training. The *ground truth* was established by an expert radiologist, who classified each *frame* as informative or non-informative according to the visualization of the anatomical structures required for each view. The binary analysis included 322 outputs; the remaining 161 were retained for uncertainty analysis. In the binary set, the system correctly identified 37.27% (60/161) of the informative *frames* and 90.68% (146/161) of the non-informative *frames*, corresponding to global sensitivity and specificity values of 37.27% and 90.68%, respectively. Performance by view was also evaluated using precision, recall, F1-score, and balanced accuracy. The F1-scores were 0.333, 0.564, 1.000, and 1.000, and the balanced accuracy values were 0.500, 0.625, 1.000, and 1.000 for the *longitudinal*, *transverse*, *oblique*, and *hepatorenal views*, respectively. These preliminary results delimit the technical feasibility of the approach on the evaluated set and support future validation with additional patients, operators, and acquisition environments.

**Keywords:** hepatic ultrasound, computer vision, informative-image identification.

## Chapter I

### Introduction

#### 1.1 Context and Problem

Ultrasound is widely used because it is portable, non-invasive, relatively low-cost, suitable for repeated examinations, and does not employ ionizing radiation [1, 2]. Its acquisition is nevertheless operator-dependent: probe position and orientation, contact, respiratory phase, rib interference, field coverage, and gain can change the anatomical content of a *frame* [8, 27]. A user with limited experience may therefore have difficulty deciding during scanning whether the current image contains the anatomical information required by a defined acquisition protocol.

Existing acquisition-assistance research includes hardware-oriented systems that constrain or robotically control the transducer, software-oriented systems that recognize views or estimate image quality, and combined tele-ultrasound workflows that connect a non-expert operator with remote expert support [2, 25, 26]. Asynchronous right-upper-quadrant tele-ultrasound has also shown that non-expert users can acquire interpretable abdominal sequences when the task is structured and reviewed by specialists [1]. These approaches establish the relevance of acquisition assistance, but they do not by themselves validate the present software or its performance on the dataset used in this work.

The problem addressed here is narrower than complete acquisition standardization. The implemented prototype identifies whether *frames* from four hepatic views satisfy expert-defined informativity criteria and returns acquisition-oriented feedback. It does not verify the physical position, angle, pressure, posture, respiratory phase, or probe contact through dedicated sensors, and it processes previously recorded videos rather than a direct live stream from an ultrasound device. Accordingly, this work does not claim to have standardized the acquisition procedure, developed a connected medical device, performed diagnosis, or completed clinical validation.

In this document, an informative image is one that satisfies the anatomical criteria established by the radiologist for the selected view. This label does not indicate normality, pathology, or guaranteed diagnostic value. The four target views are described below because they define the image content assessed by the software.

**Longitudinal view.** The acquisition procedure places the transducer parallel to the longitudinal axis of the body over the right midclavicular region, with the marker directed cephalad. Controlled inspiration can facilitate visualization of the right hepatic lobe, hepatic parenchyma, anechoic vascular structures, and diaphragm; the inferior vena cava can serve as an orientation reference when visible [31, 32]. In the annotations and code, LA denotes an anechoic lumen and is not assigned to one uniquely identified vessel.

**Transverse view.** A horizontal right-subcostal placement, with the marker directed toward the patient's right side, is used to obtain a broad transverse section of hepatic parenchyma containing visible vascular branches and adequate tissue coverage without dominant rib shadowing or loss of contact [8, 30]. The implemented branch evaluates the complete image representation and does not segment each vascular branch individually.

**Oblique view.** The transducer is positioned below the right costal margin and directed cranially and medially to produce a diagonal plane. The target contains hepatic parenchyma and vascular structures arranged along this orientation, while controlled inspiration and small angular adjustments can improve the acoustic window [8, 27]. As in the *transverse view*, the software evaluates the learned representation of the complete crop rather than explicitly identifying an individual vessel.

**Hepatorenal view.** An intercostal right-flank window, generally between the mid- and posterior-axillary lines, is used to visualize hepatic parenchyma, the right kidney, their interface, and the hepatorenal recess in the same image [1, 7]. The implemented branch evaluates the complete *frame* and does not separately segment the liver, kidney, or recess.

## 1.2 Related Work

Computer-vision methods for ultrasound acquisition support include view recognition, image-quality assessment, anatomical segmentation, and learned representation analysis. Hierarchical recognition of 11 liver scan locations reported  $F1 = 0.93$ , while Ultra-Attention reported approximately 93.2% accuracy for standard liver-section recognition [8, 12]. Self-supervised liver-view classification has further shown the value of representation learning when labeled medical data are limited [13]. These studies motivate the implemented strategies, but their anatomical targets, datasets, and validation designs differ from those used here and their numerical results are not treated as direct benchmarks.

For explicit anatomical analysis, U-Net introduced an encoder–decoder architecture with skip connections for biomedical segmentation [33]. DeepLabV3+ combines atrous spatial



pyramid pooling with a decoder for multiscale context and boundary refinement [34], whereas SegFormer uses a hierarchical transformer encoder and an efficient MLP decoder [35]. Boundary uncertainty, speckle, shadowing, and small targets remain relevant limitations in ultrasound segmentation [15, 16]. These properties motivated the controlled comparison of the three architectures for ROI, liver, and LA segmentation.

For analysis without dense masks, DINOv2 provides frozen self-supervised visual representations that can be reused as inputs to downstream models [37]. In this work, DINOv2-Small is strictly the feature extractor; the final decision is produced by a view-specific Random Forest, logistic-regression, or  $k$ -nearest-neighbor classifier. These classifiers represent established alternatives for nonlinear ensemble decisions, linear probabilistic separation, and neighborhood-based classification, respectively [38–40]. Their selection is based on the internal leave-one-patient-out comparison reported in Chapter III, not on an a priori assumption that one classifier is superior for every view.

### 1.3 Proposed System and Contributions

The proposed software uses two complementary analysis strategies. For the *longitudinal view*, DeepLabV3+ segments the ultrasound region of interest (ROI) and liver, while U-Net segments LA; post-processing and explicit morphological, intensity, texture, and border criteria then determine the image state. For the *transverse view*, *oblique view*, and *hepatorenal view*, frozen DINOv2-Small extracts one 384-feature *embedding* from each evaluated *frame*; five consecutive *embeddings* are summarized by their mean and population standard deviation into a 768-feature vector that is classified by Random Forest, logistic regression, and distance-weighted 7-NN, respectively [33, 34, 37].

The system preserves a binary informative/non-informative *ground truth* for quantitative evaluation and uses three operational feedback states in the graphical interface. Green confirms that the capture condition is satisfied; red indicates that the current image does not contain sufficient view evidence and requires repositioning; yellow indicates an uncertain or potentially informative image for which the main field is present but a fine probe adjustment is still required. In the *longitudinal view*, yellow corresponds to visible ROI and adequate liver coverage without sufficient LA-dependent evidence. In the DINOv2-Small branches, yellow corresponds to the internally calibrated probability interval between the red and green action thresholds. This third color is an abstention and guidance state, not a new expert-defined clinical class.

The engineering contribution is a traceable integration of: (i) explicit anatomical seg-

mentation and quantitative image rules for one view; (ii) frozen deep representations and view-specific classical classifiers for three views; (iii) temporal stabilization and conservative capture confirmation; and (iv) a graphical interface that processes recorded videos, updates acquired and analyzed images, provides non-diagnostic feedback, and stores *per-frame* evidence. The contribution is therefore the design, implementation, and technical evaluation of software for informative-image identification, not validation of a standardized acquisition protocol.

#### 1.4 Objectives

**General objective.** Develop and technically evaluate a computer-vision system that identifies protocol-defined informative images in four hepatic ultrasound views and exposes the result through an acquisition-assistance interface.

**Specific objectives.** Define the anatomical informativity criteria for the *longitudinal view*, *transverse view*, *oblique view*, and *hepatorenal view*; implement and compare segmentation models for the ROI, liver, and LA; implement DINOv2-Small feature extraction with view-specific classifiers; integrate deterministic decisions, temporal stabilization, and three-color feedback in a graphical interface; and evaluate the frozen system on development data and an independently acquired patient without using that patient for model or threshold selection.

#### 1.5 Document Structure

Chapter II presents the requirements, dataset, analysis alternatives, two implemented strategies, evaluation design, and graphical-interface integration. Chapter III reports model selection, independent evaluation, qualitative failures, computational performance, interface verification, and discussion. Chapter IV presents conclusions, limitations, and recommendations. The annexes provide acquisition material, data organization, code and pseudocode, reproducibility commands, interface evidence, and detailed development and independent results.

## Chapter II

### Methodology and Design

This chapter presents the method used to translate expert-defined acquisition criteria into a reproducible software workflow. It establishes the requirements and constraints, documents construction and partition of the dataset, compares the analysis alternatives, specifies the two implemented strategies, and closes with their experimental evaluation and integration into the graphical interface. The sequence follows the implemented path from a recorded source *frame* to a binary evaluation record and green, yellow, or red operational feedback.

#### 2.1 Requirements

The prototype must process recorded videos of four hepatic views, generate traceable *frame*-level decisions, and provide acquisition guidance without diagnostic claims. Tables I–III separate the required software operations, measurable quality attributes, and boundaries of the present study.

##### 2.1.1 Software Requirements

**2.1.1.1 Functional Requirements.** Table I defines the required functions and the parameter used to verify each one. Together, they trace the path from a local video and selected view to the per-*frame* result, graphical feedback, final session summary, and stored evidence.

**TABLE I**  
Functional requirements and measurement parameters

| Code | Functional requirement                                                                | Measured parameter                                                                           | Acceptance or verification                                                                                                      |
|------|---------------------------------------------------------------------------------------|----------------------------------------------------------------------------------------------|---------------------------------------------------------------------------------------------------------------------------------|
| FR1  | Process a local recorded video sequentially.                                          | Decoded <i>frames</i> and ordered indices.                                                   | At least one <i>frame</i> ; indices increase chronologically.                                                                   |
| FR2  | Select one of the four hepatic views.                                                 | View identifier and active pipeline.                                                         | Four GUI options map one-to-one to four branches.                                                                               |
| FR3  | Segment ROI, liver, and LA and apply the <i>longitudinal view</i> rule.               | Mask areas, LA features, border evidence, and raw state.                                     | Three predictions and one permitted state are returned.                                                                         |
| FR4  | Extract DINOv2-Small features and apply the view-specific classifier.                 | $5 \times 384$ <i>embeddings</i> ; 768-feature vector; RF/LR/7-NN; $P(\text{informative})$ . | The dimensions and classifier match the selected view.                                                                          |
| FR5  | Return one of three operational feedback states for every classifiable <i>frame</i> . | Green, yellow, or red user state.                                                            | Green confirms <i>capture</i> ; yellow indicates uncertainty or fine adjustment; red indicates insufficient view evidence.      |
| FR6  | Close the completed analysis and make confirmed evidence available.                   | Completion notice and downloadable informative-image archive.                                | The final notice is displayed and the archive is offered when captures exist.                                                   |
| FR7  | Display the acquired image, analyzed image, selected view, and guidance message.      | Two image panels and one status banner updated per evaluation.                               | Both panels match the current <i>frame</i> ; no area, probability, result, or capture counters are shown in the analysis panel. |
| FR8  | Save session evidence.                                                                | CSV rows, summary, candidate, and <i>capture</i> files.                                      | Summary and records exist; <i>capture</i> is conditional.                                                                       |

**2.1.1.2 Non-functional Requirements.** Table II defines the non-functional requirements. No fixed real-time threshold was imposed because the available evidence consists of measured latency and FPS rather than a validated timing requirement for continuous acquisition.

**TABLE II**  
Non-functional requirements and measurement parameters

| Code | Non-functional requirement                              | Measured parameter                                    | Acceptance or verification                                                         |
|------|---------------------------------------------------------|-------------------------------------------------------|------------------------------------------------------------------------------------|
| NFR1 | Preserve <i>per-frame</i> decision traceability.        | Index, feature/probability, state, and view.          | Each record maps to one evaluation branch.                                         |
| NFR2 | Reproduce training and inference configuration.         | Seed, checkpoint, classifier file, and configuration. | Required artifacts are identified and load correctly.                              |
| NFR3 | Report performance without assuming real time.          | Elapsed time, <i>ms/frame</i> , and FPS.              | Measured values and the tested processing branch are reported.                     |
| NFR4 | Separate image decisions from model-validation metrics. | <i>Per-frame</i> labels and aggregate metric fields.  | Dice, IoU, F1, and <i>balanced accuracy</i> are not used as direct image criteria. |
| NFR5 | Provide concise, non-diagnostic feedback.               | State-to-message mapping.                             | Every state maps to acquisition guidance only.                                     |
| NFR6 | Operate without a direct ultrasound-device connection.  | Accepted input source and acquisition interfaces.     | Local recorded-video input; live acquisition interfaces = 0.                       |

### 2.1.2 Project Constraints

The constraints in Table III delimit the interpretation of the methodology and results. In particular, the protocol specifies physical acquisition recommendations that the current software cannot sense directly, and the pilot sample does not support claims of clinical validation.

**TABLE III**  
Project and evaluation constraints

| Code | Constraint                                                                                                                                    | Bound or measured parameter                                                               | Methodological consequence                                                                                                             |
|------|-----------------------------------------------------------------------------------------------------------------------------------------------|-------------------------------------------------------------------------------------------|----------------------------------------------------------------------------------------------------------------------------------------|
| C1   | The final prototype processes recorded videos and has no direct live ultrasound input.                                                        | Number of live acquisition interfaces = 0.                                                | The system is evaluated as interactive recorded-video processing, not as an integrated real-time medical device.                       |
| C2   | The implemented scope contains four hepatic views only.                                                                                       | Number of selectable views = 4.                                                           | Conclusions are limited to the <i>longitudinal view</i> , <i>transverse view</i> , <i>oblique view</i> , and <i>hepatorenal view</i> . |
| C3   | The available development data contain few patients and one acquisition environment.                                                          | Patient identifiers and videos per split.                                                 | Splits are reported explicitly and external validity is limited.                                                                       |
| C4   | Probe frequency, exact position, angle, pressure, fasting, posture, and breath hold are protocol recommendations but are not directly sensed. | Number of dedicated position, pressure, or respiratory sensors = 0.                       | Compliance with these physical protocol items cannot be inferred from the software output alone.                                       |
| C5   | The system has no diagnostic function or multicenter clinical validation.                                                                     | Diagnostic outputs = 0; participating centers = 1.                                        | Results describe protocol-defined informativity and technical feasibility only.                                                        |
| C6   | External longitudinal videos have no independent pixel-level masks.                                                                           | External ground-truth masks = 0.                                                          | External segmentation Dice, IoU, sensitivity, and specificity cannot be calculated.                                                    |
| C7   | The acquisition pilot used one volunteer with no previous ultrasound experience after approximately 15 min of training.                       | Volunteers = 1; training time $\approx 15$ min; videos = 12; duration $\approx 3$ s each. | The pilot demonstrates execution under the defined procedure but not population-level performance.                                     |

Note: Constraints are reported as boundaries of the present work, not as functions implemented by the prototype.

### 2.1.3 Hardware Requirements

The complete prototype was executed and verified on a Windows 11 laptop with a 13th-generation Intel Core i7-13620H processor, 16 GB of RAM, and an NVIDIA GeForce RTX 4060 Laptop GPU with 8188 MiB of video memory. The verified software environment used Python 3.13.2, PyTorch 2.12.1, and CUDA 12.6. The repository, datasets, checkpoints, and generated evidence occupied approximately 15.05 GB. This is the minimum configuration demonstrated in this work; lower-resource configurations were not benchmarked and are therefore not claimed as supported.

## 2.2 Dataset Construction

Dataset construction was completed before model selection so that the development and independent-evaluation evidence remained distinct. Patients 1–3 supplied the development material; Patient 4 was reserved for the final independent evaluation.

### 2.2.1 Acquisition

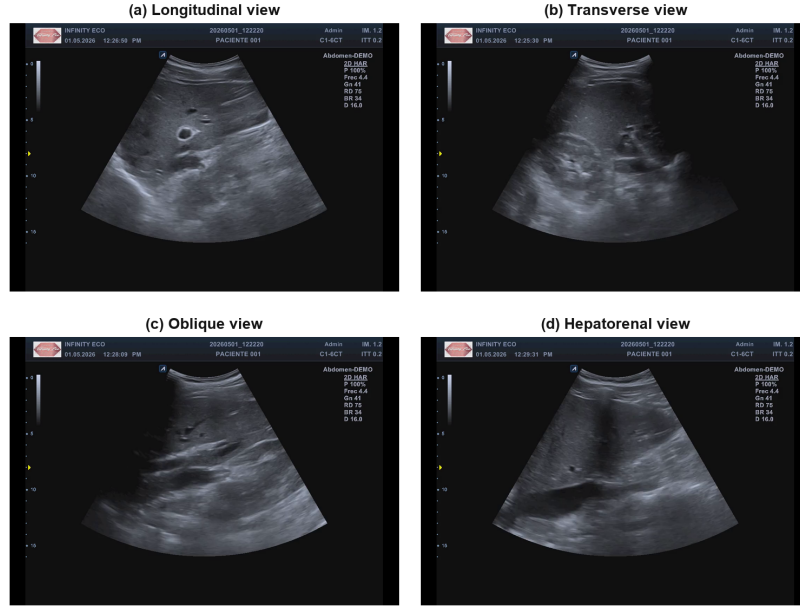
The acquisition guide used the four targets defined in the Introduction. The *transverse view* used a right-subcostal horizontal orientation, the *longitudinal view* followed the right mid-clavicular axis, the *oblique view* used cranial and medial inclination below the right costal margin, and the *hepatorenal view* used an intercostal right-flank window. Hepatic parenchyma and vascular references defined the first three targets, whereas simultaneous liver, right-kidney, and interface visualization defined the fourth. These instructions governed data collection; the software did not directly measure transducer position, angle, pressure, posture, or respiratory phase.

The final eligible dataset contained four participants. To avoid suggesting that a fifth participant was included, they are identified in this document as Patient 1, Patient 2, Patient 3, and Patient 4; the original repository codes P001, P002, P003, and P005, respectively, are retained only for file traceability. Patients 1–3 supplied the development data, whereas Patient 4 was acquired separately and reserved for independent evaluation. The complete instructions used during data collection are presented in ANNEX A.1 (*Hepatic-Ultrasound Acquisition Guide for the Four Views*).

The development set contained 4,040 source *frames*: 909 from the *longitudinal view*, 909 from the *transverse view*, 1,010 from the *oblique view*, and 1,212 from the *hepatorenal view*. All source images were  $1024 \times 768$  pixels. Patient 4 contributed twelve independent sequences of approximately 3 s, with three acquisitions of each view. These sequences were acquired by one volunteer without previous ultrasound experience after approximately 15 min of training. The exact raw-video organization is specified in ANNEX A.2 (*Repository of Raw Ultrasound Videos Used in Development and Independent Evaluation*).

Figure 1 presents one representative source *frame* for each acquisition target. The panels document the input data and should not be interpreted as independent verification of a single anatomical structure in the DINOv2 branches.

Representative source frame for each acquisition view



**Fig. 1.** Representative development-set inputs: (a) *longitudinal view*, (b) *transverse view*, (c) *oblique view*, and (d) *hepatorenal view*.

Source: Own dataset.

## 2.2.2 Labeling and Dataset Partition

For the *longitudinal-view* development data, 909 images had pixel-level ROI, liver, and LA masks in COCO-compatible form. The segmentation networks learned these masks and did not use an acquisition-quality folder as the target. For the other three views, the original development folders were *clear*, *medium*, and *blurry*: *clear* was used as the informative anchor, *blurry* as the non-informative anchor, and *medium* was excluded from classifier fitting and threshold calibration and retained only for uncertainty analysis. For Patient 4, an expert radiologist assigned an informative/non-informative label to every source *frame* according to the anatomical requirements of the corresponding view. The prepared data, annotations, and split manifests are specified in ANNEX A.3 (*Repository of Extracted Frames, Annotations, and Dataset Splits*).

Patients 1–3 were used for development, validation, model selection, and threshold derivation. Patient 4 was not used to train model weights, select segmentation architectures or classifiers, or define decision thresholds. Table IV separates source *frames* from the classifiable outputs produced by the final temporal representation.



**TABLE IV**  
Dataset organization and independent-evaluation counts

| View                     | Development<br>source frames | Patient<br>source frames | 4 Patient<br>analyzed outputs | 4 ana-<br>lyzed outputs | Label used in<br>evaluation                                                                 |
|--------------------------|------------------------------|--------------------------|-------------------------------|-------------------------|---------------------------------------------------------------------------------------------|
| <i>Longitudinal view</i> | 909                          | 303                      | 303                           |                         | Radiologist<br>informative/non-<br>informative<br>label; no inde-<br>pendent pixel<br>masks |
| <i>Transverse view</i>   | 909                          | 303                      | 60                            |                         | Radiologist<br>informative/non-<br>informative la-<br>bel                                   |
| <i>Oblique view</i>      | 1,010                        | 303                      | 60                            |                         | Radiologist<br>informative/non-<br>informative la-<br>bel                                   |
| <i>Hepatorenal view</i>  | 1,212                        | 303                      | 60                            |                         | Radiologist<br>informative/non-<br>informative la-<br>bel                                   |
| <b>Total</b>             | <b>4,040</b>                 | <b>1,212</b>             | <b>483</b>                    |                         | –                                                                                           |

Note: For each DINOv2 branch, one analyzed output is associated with the source *frame* that completes a five-*embedding* input. The 483 outputs comprise 322 binary evaluation cases and 161 uncertainty cases.

### 2.3 Conceptual Design

At the conceptual level, the system had to transform a recorded ultrasound video into a reproducible image-level decision. The literature reviewed in the Introduction supported three computational strategies directly related to the available annotations: supervised recognition of a complete hepatic view, explicit segmentation followed by measurable image rules, and frozen self-supervised representations followed by a classical classifier. Supervised hierarchical and attention-based models have recognized standardized hepatic sections [8, 12]; segmentation

provides explicit anatomical evidence [15, 33, 34]; and self-supervised methods reduce dependence on dense pixel-level labels [13, 37]. Because the *longitudinal view* had ROI, liver, and LA masks while the other views had image-level labels, the selected system combined the latter two strategies.

### 2.3.1 Analysis of Alternatives

Table V compares the three computational strategies using supervised view recognition as the datum. The symbols express compatibility with the available evidence and the stated system requirements; they are not clinical-performance measurements. The two-component alternative was selected because it preserves explicit anatomical traceability for the *longitudinal view* and uses reusable deep representations for the views without dense masks.

**TABLE V**  
Pugh matrix for computational analysis alternatives

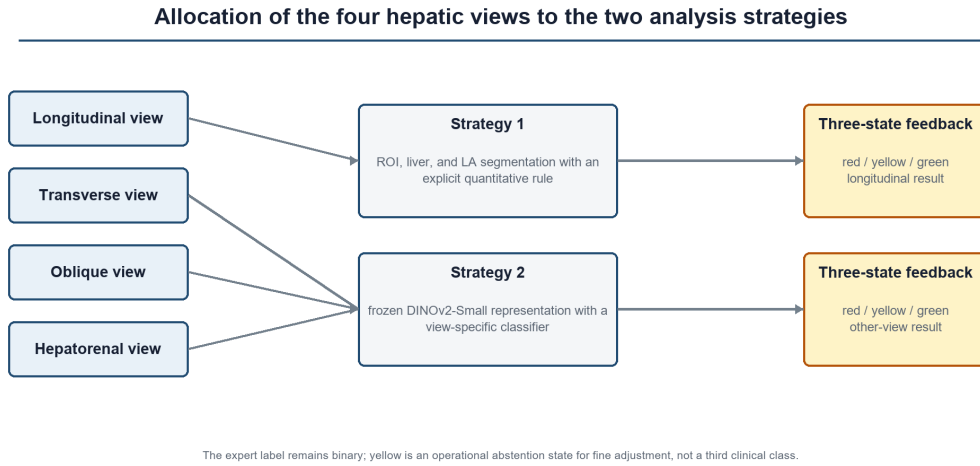
| Criterion                                    | Weight | Supervised view<br>recognition datum | Segmentation<br>and rules | Two-component<br>selected system |
|----------------------------------------------|--------|--------------------------------------|---------------------------|----------------------------------|
| Compatibility with ROI/liver/LA masks        | 5      | 0                                    | +                         | +                                |
| Compatibility with image-level view labels   | 5      | 0                                    | −                         | +                                |
| Explicit anatomical traceability             | 5      | 0                                    | +                         | +                                |
| Suitability when dense masks are unavailable | 5      | 0                                    | −                         | +                                |
| View-specific adaptation                     | 4      | 0                                    | 0                         | +                                |
| Reproducible quantitative outputs            | 4      | 0                                    | +                         | +                                |
| Weighted total                               | −      | 0                                    | +4                        | +28                              |

Note: The score follows  $S_j = \sum_i w_i s_{ij}$ , with  $s_{ij} \in \{-1, 0, +1\}$  and supervised view recognition as datum [47]. The selected alternative combines segmentation and deterministic features for the *longitudinal view* with frozen DINOv2 representations and view-specific classifiers for the remaining views. Evidence bases are supervised recognition [8, 12], segmentation [15, 33, 34], and self-supervised representation learning [13, 37].

### 2.3.2 General System Architecture

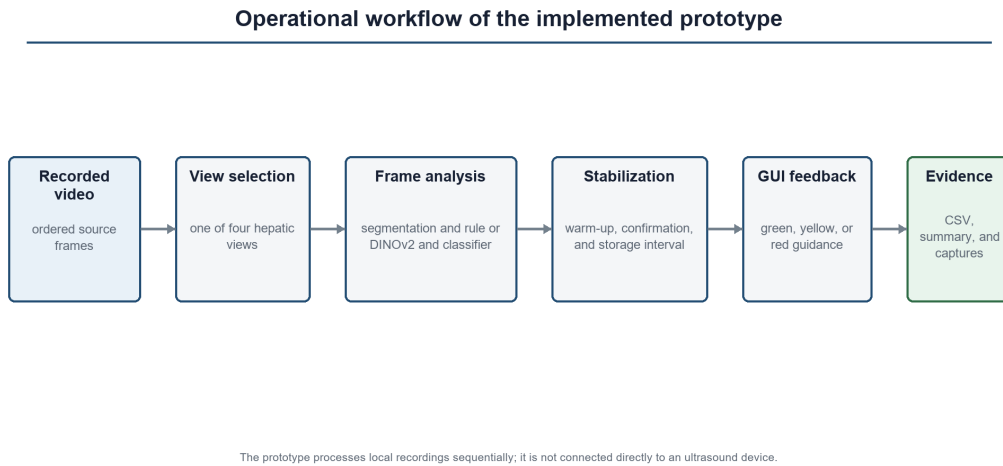
The selected architecture contains two analysis strategies connected to the same video input, graphical feedback, and evidence-writing layer. Strategy 1 processes the *longitudinal view*: it segments ROI, liver, and LA, derives the exact measurement regions, and evaluates morphological, intensity, texture, and border criteria. Strategy 2 processes the *transverse view*, *oblique view*, and *hepatorenal view*: frozen DINOv2-Small extracts one 384-feature *embedding* per evaluated *frame*, five *embeddings* are summarized into a 768-feature input, and

a view-specific classifier estimates  $P(\text{informative})$ . Both branches preserve a binary label for evaluation and map their operational result to green, yellow, or red feedback. Figure 2 identifies which view enters each strategy, while Figure 3 shows how the selected branch is integrated with temporal stabilization, user feedback, and stored evidence.



**Fig. 2.** Allocation of the four hepatic-view inputs to the two implemented analysis strategies and their three operational feedback states. The expert-defined evaluation label remains binary.

Source: Own elaboration from the implemented source code.



**Fig. 3.** Operational workflow from recorded-video input to graphical feedback and stored output evidence.

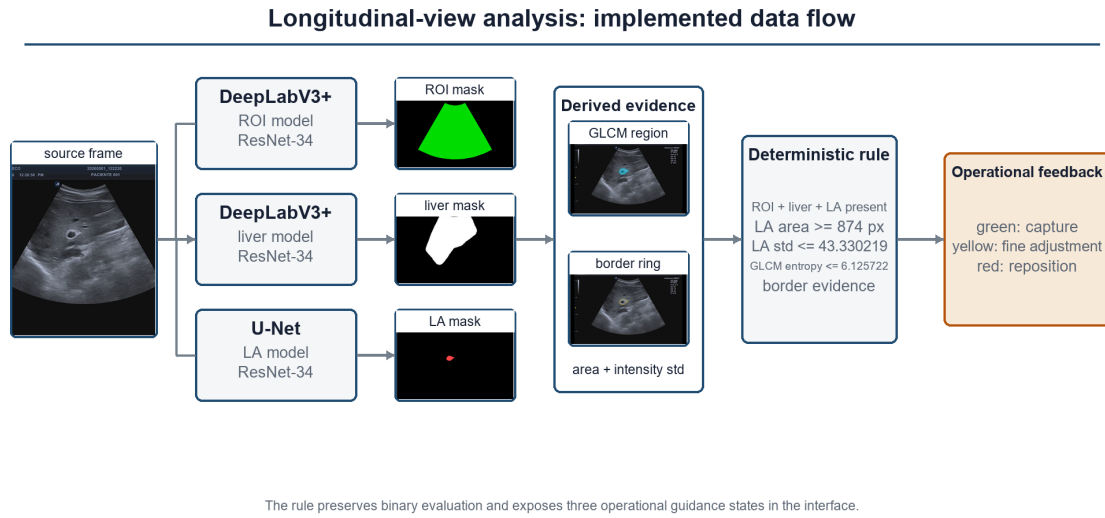
Source: Own elaboration from the implemented source code.

## 2.4 Implemented Analysis Strategies

### 2.4.1 Longitudinal View: Segmentation and Explicit Quantitative Criteria

The *longitudinal-view* branch converts one source *frame* into three masks, derives quantitative evidence from their post-processed regions, and applies a deterministic decision rule. Figure 4 fixes the order of these implemented operations from model input to operational

feedback; the following subsections define every block and threshold.

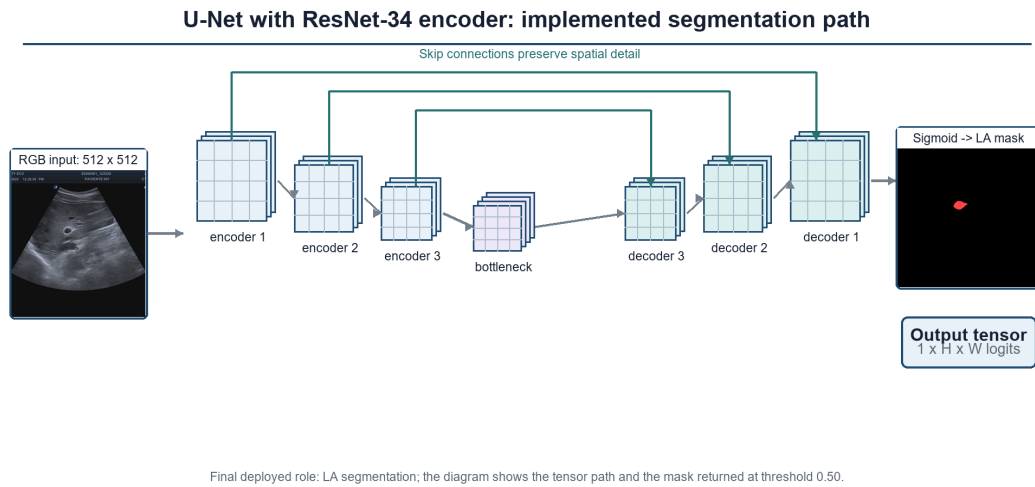


**Fig. 4.** Implemented processing sequence for the *longitudinal view*, from the *source frame* to the binary evaluation and red, yellow, or green operational feedback.

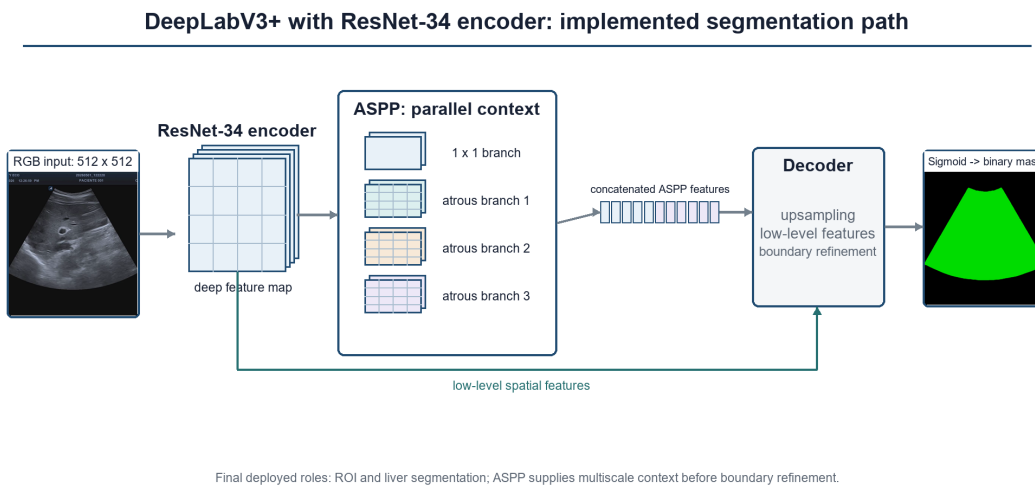
Source: Own elaboration from the implemented source code.

**2.4.1.1 Segmentation Models and Preprocessing.** The *longitudinal view* used a segmentation pipeline because annotated masks existed for ROI, liver, and LA. Three architectures were compared for binary segmentation: U-Net, DeepLabV3+, and SegFormer. U-Net was included because its skip connections preserve spatial detail in biomedical segmentation [33]. DeepLabV3+ was included because its atrous spatial pyramid pooling captures multiscale context while the decoder refines boundaries [34]. SegFormer was included as a transformer-based alternative with a hierarchical encoder and lightweight decoder [35]. Figures 5, 6, and 7 show only the implemented input-to-output path of each evaluated family and identify the encoder and decoder connections relevant to this experiment.

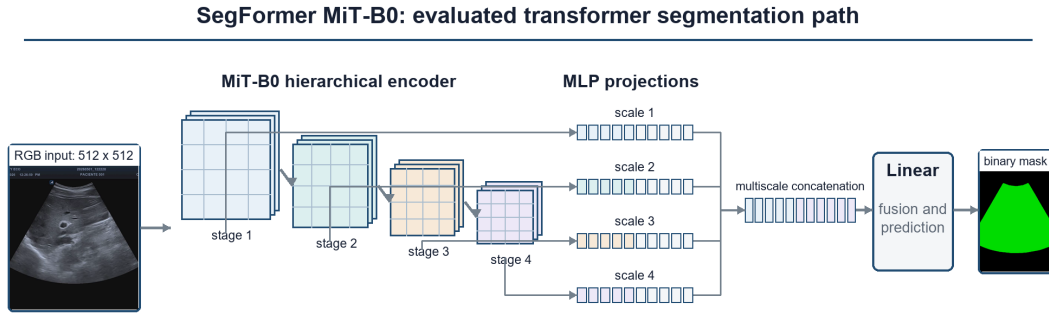
The final *longitudinal view* design used separate binary models for ROI, liver, and LA. In the annotation files and source code, LA means *lumen anecoico* (anechoic lumen). The repository does not assign this class to one uniquely identified vessel; consequently, the implemented system must not be described as explicitly segmenting the portal vein, a hepatic vein, or the inferior vena cava. LA is used as a dark visual reference whose presence, internal intensity distribution, surrounding texture, and border evidence contribute to *frame* acceptability. The final selected models were DeepLabV3+ with a ResNet-34 encoder initialized with ImageNet weights for ROI and liver, and U-Net with a ResNet-34 encoder initialized with ImageNet weights and balanced LA sampling for LA. These architecture descriptions follow the corresponding original methods [33, 34].



**Fig. 5.** U-Net input-to-output architecture evaluated for binary segmentation, including the ResNet-34 encoder, skip-connected decoder, and one-channel mask output.  
Source: Own elaboration from the implemented source code, based on U-Net [33].



**Fig. 6.** DeepLabV3+ input-to-output architecture evaluated for binary segmentation, including the ResNet-34 encoder, ASPP, decoder, and one-channel mask output.  
Source: Own elaboration from the implemented source code, based on DeepLabV3+ [34].



SegFormer was evaluated for ROI, liver, and LA; MiT-B0 multiscale features were fused by the MLP decoder, but this model was not selected.

**Fig. 7.** SegFormer input-to-output architecture evaluated for binary segmentation, including the MiT-B0 encoder, multiscale MLP decoder, and one-channel mask output.

Source: Own elaboration from the implemented source code, based on SegFormer [35].

At inference, each  $1024 \times 768$  BGR source *frame* was converted to RGB, resized to  $512 \times 512$  with OpenCV area interpolation, transposed to channels-first format, converted to floating point, and divided by 255. No denoising, contrast enhancement, ultrasound-fan crop, or ImageNet mean/standard-deviation normalization was applied to the segmentation-network input. Grayscale conversion was performed only after prediction for intensity and texture measurements. Each sigmoid output was binarized at 0.50, returned to source resolution with nearest-neighbor interpolation, cleaned by retaining the largest connected component and filling holes, and restricted hierarchically: liver to ROI and LA to liver.

**2.4.1.2 Training and Model Selection.** The *longitudinal view* dataset contained 909 images at  $1024 \times 768$  pixels. The controlled experiments used a video-grouped split with seed 42: 606 images from six videos for training, 202 images from two videos for validation, and 101 images from one video for internal testing. For LA, these partitions contained 85/521, 16/186, and 55/46 positive/empty masks, respectively. Grouping prevented *frames* from the same video from appearing in different partitions, but patients were still shared across partitions and the internal test contained one Patient 1 video. Consequently, the reported segmentation metrics represent controlled internal evaluation rather than patient-independent generalization.

Training used AdamW with a fixed learning rate of 0.001, weight decay of 0.0001, batch size 2,  $512 \times 512$  inputs, seed 42, and a maximum of 50 epochs. The loss was the sum of binary cross-entropy with logits and soft Dice loss. Early stopping used a minimum improvement of 0.0001 and patience of 10 epochs for ROI and liver and 15 for LA. ROI and liver check-

points were selected by validation Dice; LA was selected by  $Score_{LA}$ . The primary architecture comparison used transfer learning for every row under the same images, video-grouped split, preprocessing, loss, optimizer, and stopping rule: U-Net and DeepLabV3+ used ResNet-34 encoders initialized with supervised ImageNet weights, while SegFormer used the pretrained `nvidia/mit-b0` encoder, whose original checkpoint was trained on ImageNet-1k [42, 43]. All layers were fine-tuned; no learning-rate scheduler, mixed precision, gradient clipping, or staged freezing was applied. Earlier non-pretrained and augmentation runs were retained as development ablations but were not mixed with the primary comparison reported in Chapter III.

**2.4.1.3 Post-processing and Anatomical-Presence Gates.** The deployed ROI presence condition required a ratio between ROI area and full-*frame* area of at least 0.05 and rejected *frames* for which at least 80% of pixels had intensity below 15. Liver presence required an area of at least  $200(HW/786432)$  pixels, and the final state also required  $\text{liver}/\text{ROI} \geq 0.15$ . LA presence began at  $50(HW/786432)$  pixels. These presence checks are computational gates and are distinct from Dice, IoU, precision, recall, F1, and specificity, which were used only to evaluate model performance.

**2.4.1.4 Image-Decision Features and Final Rule.** The LA area was the number of nonzero pixels in the cleaned source-resolution LA mask. Its threshold was scaled with the current *frame* size:

$$A_{LA,min}^{scaled} = 874 \frac{H W}{1024 \times 768} \quad (2.1)$$

The value 874 was not obtained from an ROC curve. Script `05_definir_umbral_aceptabilidad.py` selected  $\max(20, P_5)$ , where  $P_5$  is the fifth percentile of LA area among *clear* train/validation images that contained LA and a nonempty valid region. The same script selected the 90th percentiles of the corresponding *clear* train/validation distributions for LA standard deviation and GLCM entropy, producing 43.330219 and 6.125722. The LA standard deviation was computed exactly with `numpy.std` over 8-bit grayscale intensities inside the cleaned LA mask. It was not calculated on dilated LA, ROI, or their intersection. A lower value represents a more uniform dark lumen, whereas a higher value represents greater internal intensity dispersion; this is a dataset-specific image-quality proxy rather than a physiological measurement.

The LA mask was dilated once using a  $15 \times 15$  square all-ones structuring element. The

exact GLCM region was  $ROI \cap LA_{dilated15}$ . Intensities were quantized from 256 to 32 levels using  $\lfloor 32I/256 \rfloor$ . Symmetric GLCMs were created at distance one for offsets  $(0, 1)$ ,  $(1, 0)$ ,  $(1, 1)$ , and  $(-1, 1)$ , equivalent to 0, 90, 45, and 135 degrees, and the feature values were averaged across valid offsets. GLCM-based texture analysis follows the statistical framework introduced by Haralick et al. [44]. The implementation generated contrast, entropy, homogeneity, energy, and valid-pair count; only entropy entered the final acceptance rule.

$$Contrast = \sum_{i,j} (i - j)^2 P(i, j) \quad (2.2)$$

$$Homogeneity = \sum_{i,j} \frac{P(i, j)}{1 + |i - j|}, \quad Energy = \sqrt{\sum_{i,j} P(i, j)^2} \quad (2.3)$$

$$Entropy = - \sum_{P(i,j)>0} P(i, j) \log_2 P(i, j) \quad (2.4)$$

Lower entropy indicates that the valid region contains a more concentrated distribution of gray-level pairs; higher entropy indicates more varied pair occurrences. Within this dataset, the lower-entropy side of the *clear* reference distribution was retained. Contrast, homogeneity, energy, and valid-pair count remained recorded outputs and were not hidden decision thresholds.

The border verification used an annular region around LA:

$$R_{border} = (Dilate(LA, K_{9 \times 9}) \setminus LA) \cap ROI \quad (2.5)$$

The ring used a single dilation with a  $9 \times 9$  square element. Border evidence was accepted if the bright-border branch, gradient branch, or local-contrast branch was satisfied. The bright-border branch required both of its measurements:

$$P90(R_{border}) - Median(LA) \geq 3.0 \quad (2.6)$$

$$Ratio(R_{border} \geq Median(LA) + 10) \geq 0.0375 \quad (2.7)$$

The gradient magnitude was computed on the 8-bit grayscale image with centered finite differences,  $g_x = (I_{x+1} - I_{x-1})/2$ ,  $g_y = (I_{y+1} - I_{y-1})/2$ , and  $|\nabla I| = \sqrt{g_x^2 + g_y^2}$ . The gradient



branch required:

$$P75(|\nabla I|_{R_{border}}) \geq 6.324555 \quad (2.8)$$

The local-contrast branch required:

$$P90(R_{border}) - P10(LA) \geq 35.0 \quad (2.9)$$

Script `11_ajustar_regla_aceptabilidad_borde.py` calculated these four limits as the 15th percentiles of previously base-accepted Patient 1 *clear* train/validation cases with a valid ring. In contrast, the  $15 \times 15$  dilation and  $9 \times 9$  ring kernels were fixed heuristic spatial neighborhoods; the repository contains no kernel grid search or ROC optimization. The final border expression was therefore  $(3.0 \wedge 0.0375) \vee 6.324555 \vee 35.0$ , using the corresponding measurements defined above. Figure 8 applies these exact operations to the expert-annotated ROI and LA masks of one real development *frame*. It documents the regions used to derive the thresholds; during inference, the same operations are applied to the post-processed predicted masks.

The final *longitudinal view capture* rule was:

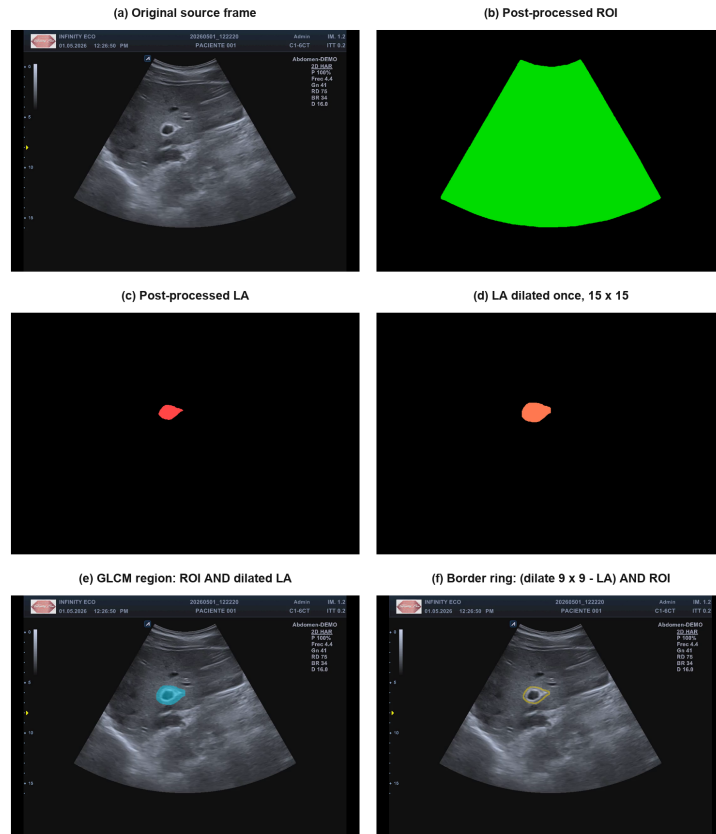
$$Capture = Rule_{base} \wedge (Border_{bright} \vee Border_{gradient} \vee Border_{contrast}) \quad (2.10)$$

where:

$$Rule_{base} = hasLA \wedge A_{LA} \geq A_{LA,min}^{scaled} \wedge \sigma_{LA} \leq 43.330219 \wedge Entropy \leq 6.125722 \quad (2.11)$$

The raw *longitudinal view* state was *no\_structure* when ROI was absent; *partial\_liver* when liver was absent or liver/ROI was below 0.15; *liver\_without\_reference* when the final LA rule failed; and *capture* otherwise. For independent binary evaluation, *capture* was informative and every other raw state was non-informative. In the GUI, raw decisions entered a separate stabilization stage: a five-decision deque selected the majority state with the most recent state as tie-breaker, a *capture* required three consecutive raw *capture* decisions, and saved captures were separated by ten evaluated *frames*. The visible mapping is red for *no\_structure* or *partial\_liver*, yellow for *liver\_without\_reference*, and green only for a confirmed *capture*. Yellow

Exact longitudinal regions used by the final decision pipeline



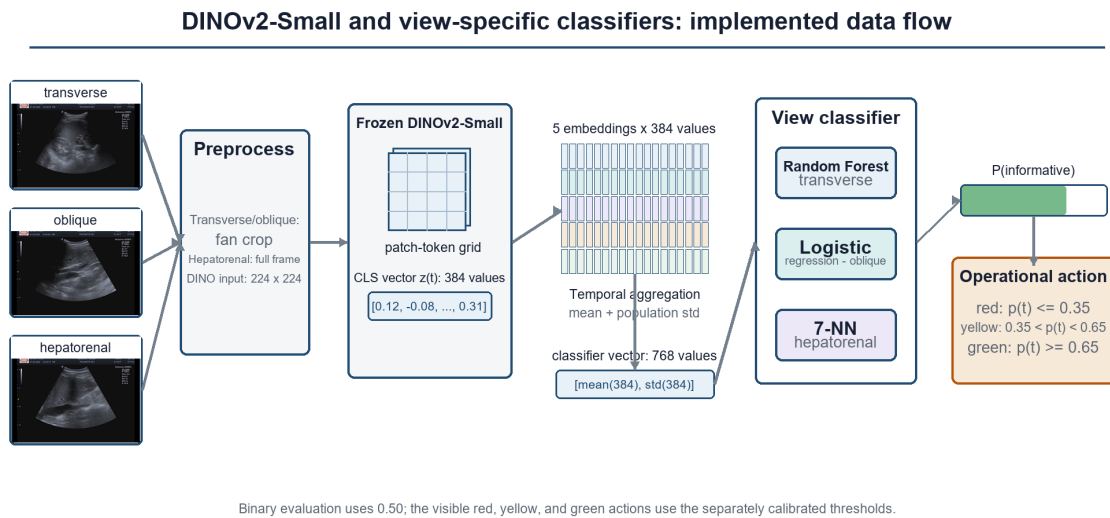
**Fig. 8.** Exact *longitudinal-view* regions used during threshold derivation on one annotated development image: (a) source *frame*; (b) ROI annotation; (c) LA annotation; (d) LA dilated once with a  $15 \times 15$  square kernel; (e)  $ROI \cap LA_{dilated15}$  used for GLCM; and (f)  $(Dilate(LA, K_{9 \times 9}) \setminus LA) \cap ROI$  used for border evidence. The final inference code applies the same operations to predicted masks. Source: Own reproducible processing of expert annotations with the implemented threshold-derivation code.

is justified because ROI and adequate liver coverage are already present, but LA is absent, too small, or fails at least one intensity, texture, or border condition. It therefore asks the user to preserve the hepatic window and make a fine inclination adjustment rather than discard the view completely. Raw states remain in the technical CSV for traceability; the yellow state is operational guidance and does not alter the binary expert label.

The scripts that implement segmentation, post-processing, feature extraction, threshold application, and temporal confirmation for this strategy are described in ANNEX B.1 (*Source Code for Longitudinal-View Segmentation and Deterministic Decision*).

### 2.4.2 Transverse View, Oblique View, and Hepatorenal View: DINOv2-Small and View-Specific Classifiers

These three branches use the same frozen feature extractor and temporal representation but retain a different classical classifier for each view. Figure 9 summarizes the implemented path from the view-dependent input crop to probability estimation and the three operational actions; the following subsections specify preprocessing, vector construction, classifier parameters, and temporal confirmation.



**Fig. 9.** Implemented DINOv2-Small processing sequence from one source *frame* to a view-specific probability and red, yellow, or green operational feedback. Binary evaluation uses the separate 0.50 boundary.

Source: Own elaboration from the implemented source code.

**2.4.2.1 Feature Extraction and Preprocessing.** For these three views, the frozen facebook/dinov2-small model was used only as a feature extractor, following the self-supervised representation-learning approach of DINOv2 [37]. Its processor converted input to RGB, resized the shortest edge to 256, center-cropped to  $224 \times 224$ , rescaled by  $1/255$ , and normalized

with mean (0.485, 0.456, 0.406) and standard deviation (0.229, 0.224, 0.225). The frozen CLS token produced a 384-dimensional *embedding* per evaluated *frame*. The *transverse view* and *oblique view* used the deterministic ultrasound-fan crop, whereas the *hepatorenal view* used the full *frame*.

**2.4.2.2 Temporal Aggregation.** A window of five *embeddings* was summarized by mean and population standard deviation, producing a 768-dimensional classifier input:

$$\phi_t = [\mu(z_{t-4:t}), \sigma(z_{t-4:t})] \quad (2.12)$$

Offline training used non-overlapping five-*frame* windows within each video, while inference maintained a sliding deque of the five most recent *embeddings*. The first four evaluated *frames* returned *warming\_up* because the 768-feature representation was not yet complete.

**2.4.2.3 View-Specific Classifiers.** The selected *transverse view* classifier was a Random Forest with 300 trees, minimum leaf size 2, balanced-subsample class weighting, seed 42, and all CPU cores. The *oblique view* used `StandardScaler` followed by logistic regression with  $C = 1.0$ , 5000 maximum iterations, balanced class weighting, and seed 42. The *hepatorenal view* used `StandardScaler` followed by seven-nearest-neighbor classification with distance weighting. All three classifiers returned `predict_proba`;  $p_t$  was the probability assigned to the informative development anchor. The original *medium* videos were excluded from fitting and threshold calibration.

**2.4.2.4 Decision Criteria and Temporal Confirmation.** After the first four *frames* initialize the five-*embedding* representation, the binary anchor class used for model comparison and independent evaluation was defined by:

$$Class(p_t) = \begin{cases} non-informative, & p_t < 0.50 \\ informative, & p_t \geq 0.50 \end{cases} \quad (2.13)$$

The 0.50 boundary is the binary decision used for independent evaluation and was not optimized on Patient 4. Function `calibrate_abstention_thresholds()` separately searched internal development probabilities from 0.01 to 0.99 in steps of 0.01. It required at least 0.90 action precision and then maximized class coverage, producing 0.35 for *adjust* and 0.65 for *capture*. The operational mapping is red when  $p_t \leq 0.35$ , yellow when  $0.35 < p_t < 0.65$ , and green when  $p_t \geq 0.65$  after temporal confirmation. Yellow is appropriate because the

classifier does not provide sufficient internal evidence for either a confident rejection or capture; the message asks the user to maintain the anatomical window and make a fine positional or angular adjustment. Three consecutive *capture* actions are required before green feedback and storage confirmation, and confirmed captures are separated by ten evaluated *frames*. The first four evaluations remain neutral while the five-*embedding* vector is initialized. These interface actions do not redefine the binary *ground truth*. The implementation and selected classifier bundles are described in ANNEX B.2 (*Source Code for DINOv2-Small Feature Extraction and View-Specific Classification*).

## 2.5 Experimental Configuration and Evaluation Metrics

The DINOv2 development subset contained 3,131 source *frames* from 31 videos of Patients 1–3; a stride of five produced 651 development *embeddings*. DINOv2-Small and DINOv2-Base, full-*frame* and deterministic fan-crop variants, individual-*frame* and five-*embedding* representations, and four classifiers were compared through leave-one-patient-out validation. Random Forest, logistic regression, RBF-SVM, and  $k$ -NN were evaluated for every view [38–41]. The primary selection criterion was mean video macro-F1 over the three held-out-patient folds; sample macro-F1 and deployment cost were supporting criteria. Patient 4 and the uncertainty subset were excluded from model selection and fitting.

Table VI records the alternatives that were actually evaluated. This table is distinct from the conceptual Pugh matrix: the Pugh matrix compares system concepts, whereas Table VI compares implementations within the selected concept.

**TABLE VI**  
Experimentally evaluated model and classifier alternatives

| Task                  | Alternatives evaluated                                                                               | Selection evidence                                                                                    | Selected implementation                                               |
|-----------------------|------------------------------------------------------------------------------------------------------|-------------------------------------------------------------------------------------------------------|-----------------------------------------------------------------------|
| ROI segmentation      | U-Net [33], DeepLabV3+ [34], SegFormer [35]                                                          | Validation Dice; IoU and latency as supporting evidence                                               | DeepLabV3+, ResNet-34, ImageNet                                       |
| Liver segmentation    | U-Net [33], DeepLabV3+ [34], SegFormer [35], all with transfer initialization                        | Validation Dice, followed by frozen internal test                                                     | DeepLabV3+, ResNet-34, ImageNet                                       |
| LA segmentation       | U-Net [33], DeepLabV3+ [34], SegFormer [35]; balanced sampling, resize, and augmentation variants    | $Score_{LA} = Dice_{positive} - FP_{empty}$ on validation                                             | U-Net, ResNet-34, ImageNet, balanced LA                               |
| DINOv2 representation | Small/full, Small/fan crop, Base/fan crop; single <i>frame</i> and five- <i>embedding</i> input [37] | Mean video macro-F1 under leave-one-patient-out validation; sample F1 and cost as supporting evidence | DINOv2-Small and five- <i>embedding</i> input                         |
| Binary classifier     | Random Forest, logistic regression, RBF-SVM, and $k$ -NN [38–41] for each DINOv2-Small view          | View-specific leave-one-patient-out comparison                                                        | RF/ <i>transverse</i> ; LR/ <i>oblique</i> ; 7-NN/ <i>hepatorenal</i> |

Note: Patient 4 remained excluded from all selection decisions.

Image-decision criteria and model-validation metrics were kept separate. Area, intensity standard deviation, GLCM entropy, border evidence, and classifier probability determine an individual image result. Dice, IoU, precision, recall, specificity, macro-F1, and *balanced accuracy* summarize performance over a labeled set and do not enter the decision for one image. For pixel-level segmentation, the reported metrics were calculated as:

$$Dice = \frac{2TP}{2TP + FP + FN}, \quad IoU = \frac{TP}{TP + FP + FN} \quad (2.14)$$

$$Precision = \frac{TP}{TP + FP}, \quad Recall = \frac{TP}{TP + FN}, \quad Specificity = \frac{TN}{TN + FP} \quad (2.15)$$

$$F1 = \frac{2 Precision Recall}{Precision + Recall} \quad (2.16)$$

For LA, positive-mask overlap was supplemented by the empty-ground-truth false-positive

rate because several images contained no LA annotation:

$$FP_{empty} = \frac{N_{empty, pred>0}}{N_{empty}}, \quad Score_{LA} = Dice_{positive} - FP_{empty} \quad (2.17)$$

**TABLE VII**  
Deployed processing parameters by ultrasound view

| Parameter               | <i>Longitudinal view</i>                      | <i>Transverse view</i>                                                                                        | <i>Oblique view</i>         | <i>Hepatorenal view</i>       |
|-------------------------|-----------------------------------------------|---------------------------------------------------------------------------------------------------------------|-----------------------------|-------------------------------|
| Model                   | Three binary segmenters                       | Frozen DINOv2-Small CLS                                                                                       | Frozen DINOv2-Small CLS     | Frozen DINOv2-Small CLS       |
| Input                   | RGB full resize, $512^2$                      | Fan crop, processor $224^2$                                                                                   | Fan crop, processor $224^2$ | Full frame, processor $224^2$ |
| Classifier/rule         | Anatomical and image-feature rule             | Random Forest                                                                                                 | Logistic regression         | Distance-weighted k-NN        |
| Score                   | Boolean conditions                            | $P(informative)$                                                                                              | $P(informative)$            | $P(informative)$              |
| Thresholds              | Area, std, entropy, border; raw-state mapping | Evaluation: $P(informative) \geq 0.50$ ; actions: red $\leq 0.35$ , yellow $(0.35, 0.65)$ , green $\geq 0.65$ | Same                        | Same                          |
| Frame context           | GUI: five-decision majority, confirmation 3   | Five-embedding input; warm-up 4; confirmation 3                                                               | Same                        | Same                          |
| Stored-capture interval | 10 evaluated frames                           | 10 evaluated frames                                                                                           | Same                        | Same                          |

Note: The table separates the deployed decision rule or classifier from the model used to obtain masks or representations.

Table VII makes the two computational components explicit: segmentation and deterministic features for the *longitudinal view*, and frozen DINOv2 representations followed by three different classical classifiers for the other views. It also separates the *five-embedding* classifier input from the binary decision assigned to each classifiable *frame*. The inference benchmark was calculated as:

$$ms/frame = 1000 \frac{T_{elapsed}}{N_{frames}}, \quad FPS = \frac{N_{frames}}{T_{elapsed}} \quad (2.18)$$

For the serial *longitudinal view* pipeline, the expected per-frame time was approximated by the sum of the three model times and post-processing time:

$$t_{pipeline} = t_{ROI} + t_{Liver} + t_{LA} + t_{post}, \quad FPS_{pipeline} = \frac{1000}{t_{pipeline,ms}} \quad (2.19)$$

The exact commands and parameter-to-source mapping needed to reproduce these measurements and decisions are presented in ANNEX B.3 (*Reproduction Commands and Parameter-to-Code Traceability*).

## 2.6 GUI Integration and Stored Evidence

The graphical interface integrates both analysis strategies without direct connection to an ultrasound machine or camera. It accepts a local recorded video or uploaded video file, requires selection of one of the four views, and permits configuration of processing stride, segmentation threshold, maximum evaluated *frames*, screen-update interval, capture-storage interval, and CPU execution. During processing, two fixed panels are updated in place: the acquired *frame* and its analyzed output. Once a classifiable result is available, the visible feedback is green for a confirmed informative capture, yellow for a potentially informative or uncertain image that requires fine adjustment, and red when the view evidence is insufficient and repositioning is required. The warm-up stage is shown neutrally as analysis in progress. The analysis panel does not expose the *frame* index, LA area, probability, raw result, capture count, or timing statistics. At completion, it displays an explicit end-of-analysis notice and offers a ZIP download containing the confirmed informative images when such captures exist.

Each session stores a per-frame CSV, a session summary, conditional *capture* images, and the best confirmed or best available candidate. Technical fields preserve ROI, liver, LA, intensity, texture, border, probability, timing, and state evidence according to the active branch. The user-facing message remains limited to acquisition guidance and does not report diagnosis, disease, or anatomical normality.

The GUI execution command and the structure of its stored outputs are presented in ANNEX C.1 (*Graphical Interface Workflow, Stored Outputs, and Downloadable Captures*). The end-to-end checks applied to the integrated implementation are listed in ANNEX C.2 (*End-to-End Verification Protocol for the Integrated System*).



## Chapter III

### Results

This chapter reports the evidence in the order in which it supports the final system: selection of the three segmentation models, selection of the three DINOv2-Small representations and view-specific classifiers, independent evaluation on Patient 4, qualitative error analysis, computational performance, and graphical-interface verification. Internal results justify component selection; only the Patient 4 results evaluate the complete frozen system on an independently acquired subject.

#### 3.1 Segmentation-Model Selection

The controlled internal test contained 101 *longitudinal-view* images. U-Net, DeepLabV3+, and SegFormer were compared under the same  $512 \times 512$ , no-augmentation, video-grouped configuration and with transfer initialization for every architecture. Model choice used validation data only; the internal test was read after selection. Results are reported separately for ROI, liver, and LA because their class balance and error modes were different.

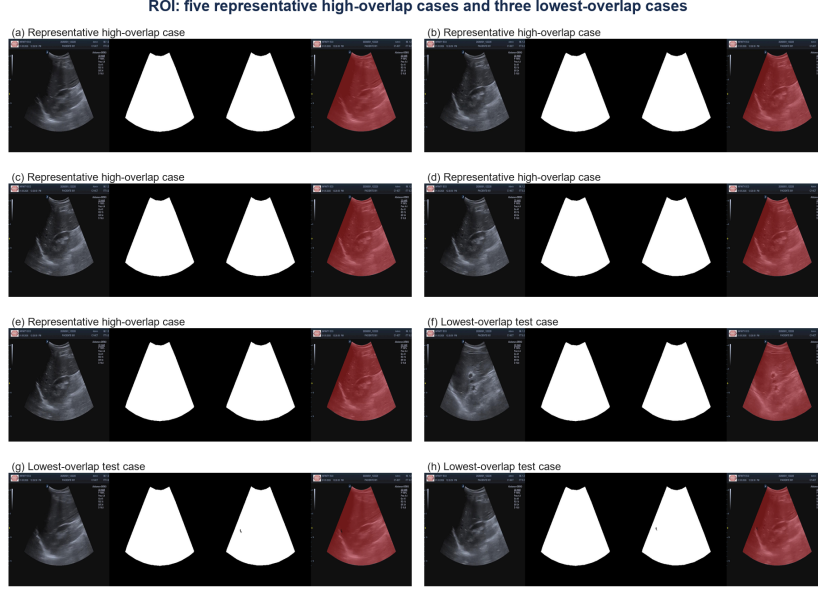
##### 3.1.1 ROI Segmentation

DeepLabV3+ obtained the highest validation Dice for ROI and was therefore selected with its ImageNet-initialized ResNet-34 encoder. Table VIII shows that the three transfer-initialized architectures were close on the internal test, with the selected model reaching Dice 0.9929 and IoU 0.9859. The qualitative examples in Figure 10 show stable fan-region coverage in panels (a)–(e); panels (f)–(h) are the three lowest-overlap test cases, where the residual error is concentrated at the fan boundary rather than in a completely missing ROI.

**TABLE VIII**  
Transfer-learning comparison for ROI segmentation on 101 internal-test images

| Architecture    | Validation Dice | Test Dice     | Test IoU      | Test recall |
|-----------------|-----------------|---------------|---------------|-------------|
| U-Net [33]      | 0.9927          | 0.9922        | 0.9846        | 0.9851      |
| DeepLabV3+ [34] | <b>0.9948</b>   | <b>0.9929</b> | <b>0.9859</b> | 0.9865      |
| SegFormer [35]  | 0.9864          | 0.9926        | 0.9853        | 0.9898      |

Note: U-Net and DeepLabV3+ used ImageNet-initialized ResNet-34 encoders; SegFormer used the ImageNet-1k-pretrained MiT-B0 encoder. All layers were fine-tuned, and DeepLabV3+ was selected by validation Dice.



**Fig. 10.** ROI qualitative test evidence: panels (a)–(e) are five representative high-overlap outputs; panels (f)–(h) are the three lowest-overlap outputs. Each panel preserves the original image, ground-truth mask, predicted mask, and overlay generated by the evaluation pipeline.

Source: Own internal test evidence.

### 3.1.2 Liver Segmentation

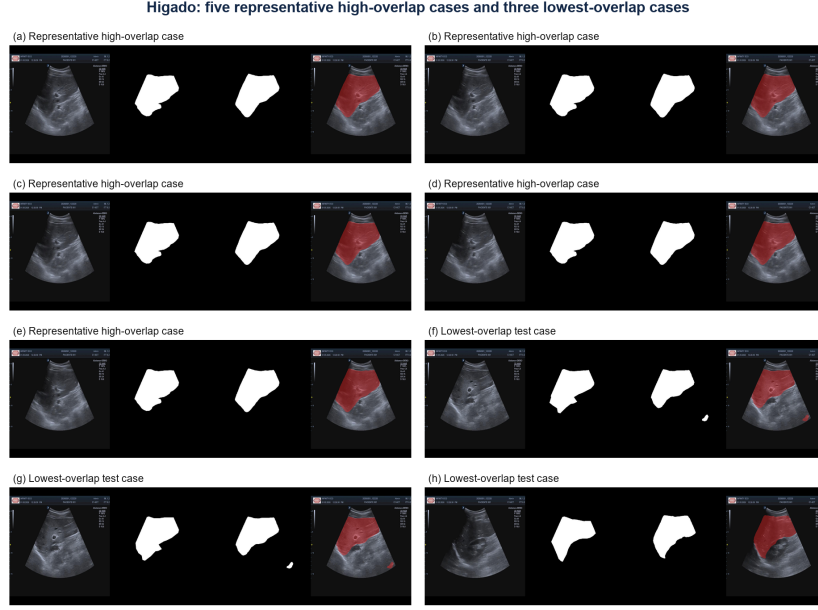
DeepLabV3+ also led the controlled transfer-learning comparison for liver segmentation and was therefore frozen as the final liver segmenter. Table IX compares the three architectures under the same transfer-learning condition, and Figure 11 shows that the best cases follow the parenchymal contour, while the lowest-overlap cases display boundary displacement and incomplete inferior coverage.

**TABLE IX**

Transfer-learning comparison for liver segmentation on 101 internal-test images

| Configuration               | Validation Dice | Test Dice     | Test IoU      | Test recall |
|-----------------------------|-----------------|---------------|---------------|-------------|
| U-Net [33], ImageNet        | 0.9257          | 0.9404        | 0.8879        | 0.9532      |
| DeepLabV3+ [34], ImageNet   | <b>0.9328</b>   | <b>0.9416</b> | <b>0.8902</b> | 0.9439      |
| SegFormer [35], ImageNet-1k | 0.8949          | 0.9235        | 0.8611        | 0.9695      |

Note: All three rows used transfer initialization and full fine-tuning under the same data split and training configuration. DeepLabV3+ was selected by validation Dice.



**Fig. 11.** Liver qualitative test evidence: panels (a)–(e) are five representative high-overlap outputs; panels (f)–(h) are the three lowest-overlap outputs.  
Source: Own internal test evidence.

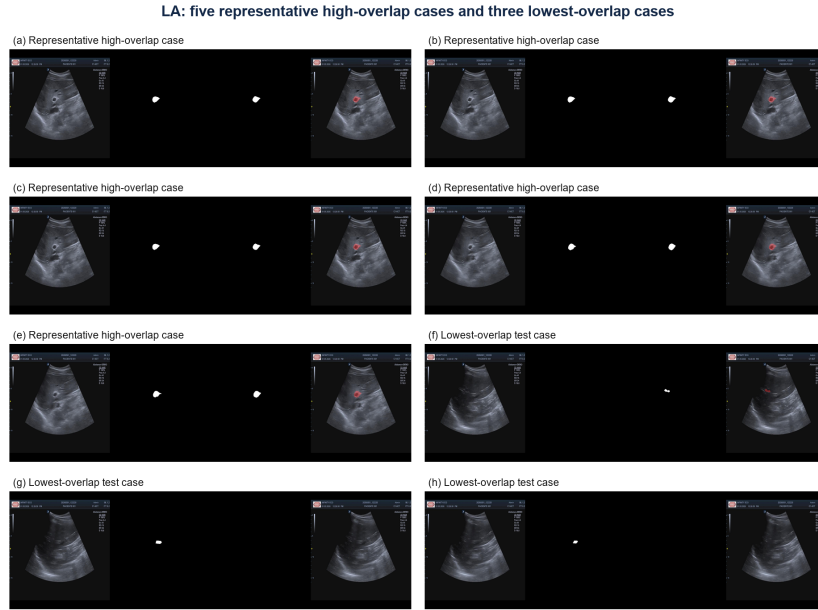
### 3.1.3 LA Segmentation

LA required a different selection rule because 46 of the 101 test masks were empty. The global Dice could be increased by predicting little or no LA; consequently, validation used  $Score_{LA} = Dice_{positive} - FP_{empty}$ . As shown in Table X, ImageNet-initialized U-Net produced the highest validation score, test Dice 0.8425, positive-case Dice 0.8198, and false-positive masks in 6 of 46 empty cases (13.04%). Figure 12 illustrates the remaining difficulty: the failures involve small, shifted, or fragmented predictions around a structure occupying few pixels.

**TABLE X**  
Transfer-learning comparison for LA segmentation on 101 internal-test images

| Configuration               | Validation<br>$Score_{LA}$ | Test Dice     | Positive Dice | $FP_{empty}$         |
|-----------------------------|----------------------------|---------------|---------------|----------------------|
| U-Net [33], ImageNet        | <b>0.3226</b>              | <b>0.8425</b> | 0.8198        | <b>13.04% (6/46)</b> |
| DeepLabV3+ [34], ImageNet   | 0.2675                     | 0.8114        | <b>0.8355</b> | 21.74%<br>(10/46)    |
| SegFormer [35], ImageNet-1k | 0.1893                     | 0.7488        | 0.7386        | 23.91%<br>(11/46)    |

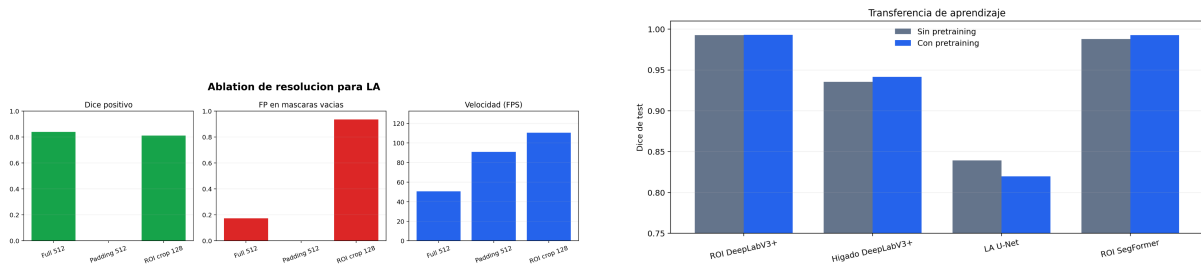
Note: All rows used transfer initialization, balanced LA sampling, and full fine-tuning under the same data split and training configuration. The final U-Net was selected by validation  $Score_{LA}$ , not by the internal-test values in the last three columns.



**Fig. 12.** LA qualitative test evidence: panels (a)–(e) are five representative high-overlap outputs; panels (f)–(h) are the three lowest-overlap outputs.

Source: Own internal test evidence.

The resolution ablation and transfer-learning comparisons support these selections. The 128-pixel ROI crop increased speed but produced excessive false positives on empty LA masks, so the final branch retained the full  $512 \times 512$  resize. ImageNet initialization improved the final liver result and reduced LA empty-mask false positives from 17.39% to 13.04%, although it did not improve every LA overlap statistic. Figure 13 summarizes these two auxiliary experiments in one compact panel.



**Fig. 13.** Auxiliary segmentation experiments: resolution ablation (left) and transfer-learning comparison (right).

Source: Own experimental results.

The strong ROI and liver overlap is consistent with the use of encoder–decoder and multiscale segmentation architectures, while the LA failures reflect the known difficulty of small ultrasound targets and boundary uncertainty [15, 16, 33, 34]. These citations motivate the architectures and interpretation; their datasets are different, so their published values are not treated as direct numerical benchmarks for this experiment. The scripts and archived outputs supporting the segmentation comparisons are presented in ANNEX B.1 (*Source Code for Longitudinal-View Segmentation and Deterministic Decision*).

### 3.2 DINOv2-Small Classifier Selection

#### 3.2.1 Leave-One-Patient-Out Comparison

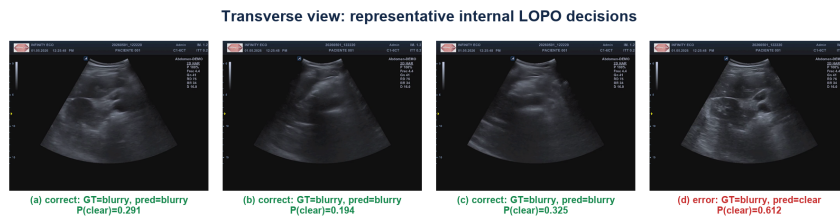
The DINOv2 alternatives were compared separately for each view through leave-one-patient-out validation over Patients 1–3. Table XI fixes the representation selected for each view and changes only the classifier, preventing a classifier effect from being confused with a different crop, backbone, or temporal representation. The primary value is macro-F1 after aggregating the samples of each held-out video; sample macro-F1 is included because a perfect video-level value can still contain individual five-embedding errors.

**TABLE XI**  
View-specific classifier comparison with the selected DINOv2 representation

| View                    | Classifier               | Video macro-F1 | Video balanced accuracy | Sample macro-F1 |
|-------------------------|--------------------------|----------------|-------------------------|-----------------|
| <i>Transverse view</i>  | <i>k</i> -NN [40]        | 0.778          | 0.833                   | 0.771           |
|                         | Logistic regression [39] | 1.000          | 1.000                   | 0.915           |
|                         | Random Forest [38]       | <b>1.000</b>   | <b>1.000</b>            | <b>0.958</b>    |
|                         | RBF-SVM [41]             | 1.000          | 1.000                   | 0.874           |
| <i>Oblique view</i>     | <i>k</i> -NN [40]        | 0.667          | 0.750                   | 0.605           |
|                         | Logistic regression [39] | <b>1.000</b>   | <b>1.000</b>            | <b>0.958</b>    |
|                         | Random Forest [38]       | 0.778          | 0.833                   | 0.665           |
|                         | RBF-SVM [41]             | 0.556          | 0.667                   | 0.526           |
| <i>Hepatorenal view</i> | <i>k</i> -NN [40]        | <b>1.000</b>   | <b>1.000</b>            | <b>0.860</b>    |
|                         | Logistic regression [39] | 0.578          | 0.667                   | 0.686           |
|                         | Random Forest [38]       | 0.356          | 0.500                   | 0.742           |
|                         | RBF-SVM [41]             | 0.356          | 0.500                   | 0.429           |

Note: Bold rows are the deployed classifiers. Every row uses frozen DINOv2-Small [37]; the *transverse* and *oblique* rows use a fan crop and five *embeddings*, while *hepatorenal* uses the full *frame* and five *embeddings*.

For the *transverse view*, logistic regression, Random Forest, and RBF-SVM tied at 1.000 video macro-F1. Random Forest was selected because its sample macro-F1 was the highest of the tied configurations. Figure 14 shows three correct internal samples and the single observed error for this selected configuration; the error assigned a non-informative sample to the informative class with  $P(\text{informative}) = 0.612$ .

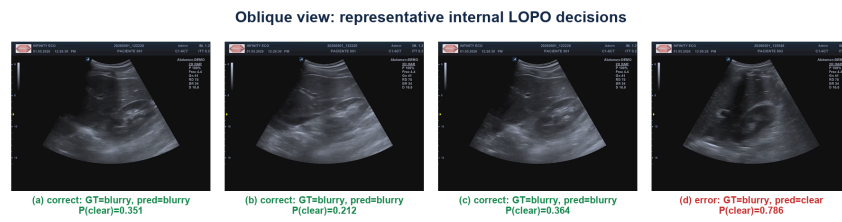


**Fig. 14.** Selected *transverse-view* internal LOPO examples: panels (a)–(c) are correct and panel (d) is the only sample-level error found in the selected archived predictions.

Source: Own leave-one-patient-out results.

For the *oblique view*, logistic regression was the only classifier to reach 1.000 video macro-F1 and *balanced accuracy* under the selected representation. Figure 15 again shows

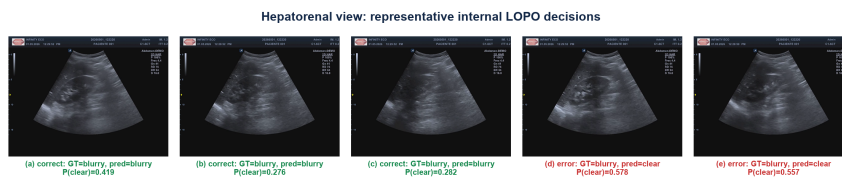
three correct samples and the only archived sample-level error, a non-informative sample with  $P(\text{informative}) = 0.786$ .



**Fig. 15.** Selected *oblique-view* internal LOPO examples: three correct samples and the single observed sample-level error.

Source: Own leave-one-patient-out results.

For the *hepatorenal view*, distance-weighted 7-NN was the only classifier with 1.000 video macro-F1 and *balanced accuracy*. Its sample macro-F1 was 0.860, confirming that video aggregation concealed some local errors. Figure 16 shows three correct samples and two of the four observed sample-level errors; all four errors remain counted in the archived CSV.



**Fig. 16.** Selected *hepatorenal-view* internal LOPO examples: three correct samples and two representative sample-level errors.

Source: Own leave-one-patient-out results.

These internal values are selection evidence, not independent performance estimates. Each fold held out one of the three development patients, and the perfect video-level values were obtained from few videos. DINOv2-Small supplied the frozen representation, while the decision came from the view-specific classifier [37]. The comparison scripts and archived output manifests are presented in ANNEX B.2 (*Source Code for DINOv2-Small Feature Extraction and View-Specific Classification*).

### 3.3 Independent Evaluation of the Integrated System

Patient 4 (repository code P005) did not participate in training, model selection, or threshold derivation. The independent evaluation comprised twelve sequences of approximately 3 s, with three acquisitions of each hepatic view. An expert radiologist assigned the informative/non-informative *ground truth* for every source *frame*. The final anchor-based binary analysis comprised 322 outputs: 202 from the *longitudinal view* and 40 classifiable outputs from

each DINOv2-Small branch. A further 161 outputs from the predefined intermediate-quality acquisitions were retained for uncertainty analysis and were not included in the binary metrics. The red/yellow/green interface mapping was applied after the algorithmic decision; it was not evaluated as a three-class clinical target because the expert *ground truth* was binary.

Table XII is the central independent-evaluation table. All performance metrics are reported as percentages to avoid mixing proportions and percentages in the same column. Precision and recall refer to the informative class; macro-F1 averages the two class-specific F1-scores.

**TABLE XII**

Independent Patient 4 performance, including correct informative and non-informative counts by view

| View                | Informative detected<br>/ informative GT | Non-informative rejected<br>/ non-informative GT | Correct<br>/ total     | Precision    | Recall       | Macro-F1     | Balanced<br>accuracy |
|---------------------|------------------------------------------|--------------------------------------------------|------------------------|--------------|--------------|--------------|----------------------|
| <i>Longitudinal</i> | 0/101 (0.0%)                             | 101/101 (100.0%)                                 | 101/202 (50.0%)        | 0.0%         | 0.0%         | 33.3%        | 50.0%                |
| <i>Transverse</i>   | 20/20 (100.0%)                           | 5/20 (25.0%)                                     | 25/40 (62.5%)          | 57.1%        | 100.0%       | 56.4%        | 62.5%                |
| <i>Oblique</i>      | 20/20 (100.0%)                           | 20/20 (100.0%)                                   | 40/40 (100.0%)         | 100.0%       | 100.0%       | 100.0%       | 100.0%               |
| <i>Hepatorenal</i>  | 20/20 (100.0%)                           | 20/20 (100.0%)                                   | 40/40 (100.0%)         | 100.0%       | 100.0%       | 100.0%       | 100.0%               |
| <b>Global</b>       | <b>60/161 (37.3%)</b>                    | <b>146/161 (90.7%)</b>                           | <b>206/322 (64.0%)</b> | <b>80.0%</b> | <b>37.3%</b> | <b>61.2%</b> | <b>64.0%</b>         |

Note: GT denotes the expert-radiologist *ground truth*. The first two result columns show how many informative and non-informative outputs were correctly identified relative to the number evaluated in each class. Patient 4 was excluded from all development and calibration procedures.

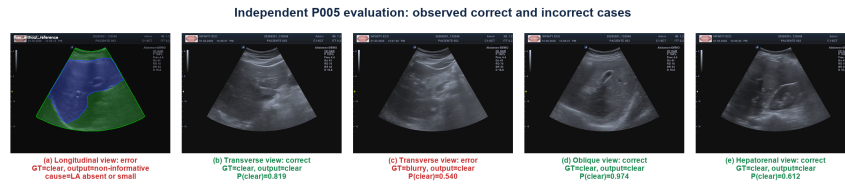
Globally, the system correctly classified 206 of 322 outputs. Specificity was 90.7% and sensitivity was 37.3%. The combination indicates conservative acceptance: most non-informative outputs were rejected, but many outputs labeled informative by the radiologist were not accepted. The exact TP, TN, FP, and FN counts are presented in ANNEX D.2 (*Patient 4 Independent Confusion Counts by Hepatic View*).

### 3.4 Qualitative Analysis of Correct Cases and Failures

The largest independent error occurred in the *longitudinal view*, which rejected all 101 informative outputs; this agrees with the observed LA bottleneck. The *transverse view* detected all 20 informative outputs but misclassified 15 of 20 non-informative outputs. In contrast, the *oblique view* and *hepatorenal view* correctly classified their 40 binary outputs. Each of these two view subsets was obtained from only one informative and one non-informative independent sequence, so the 100.0% values are preliminary technical results rather than evidence of generalization.



Figure 17 makes these outcomes concrete. Panels (b), (d), and (e) are representative source *frames* from correctly classified informative videos. Panel (c) represents the non-informative video of the *transverse view* incorrectly assigned to the informative class, with  $P(\text{informative}) = 0.540$ . Panel (a) is a false negative of the *longitudinal view*: ROI and liver were detected, but LA was absent or below the area rule, so the complete deterministic condition returned non-informative. The probabilities shown are the archived video-level probabilities; the images are representative source *frames* from those videos.



**Fig. 17.** Representative independent Patient 4 cases: (a) longitudinal false negative caused by insufficient LA; (b) correct informative *transverse-view* case; (c) non-informative *transverse-view* case misclassified as informative; (d) correct informative *oblique-view* case; and (e) correct informative *hepatorenal-view* case.

Source: Own independent-evaluation evidence.

The per-development-patient outputs are retained in ANNEX D.1 (*Development-Patient Outputs for Model Selection*), while the independent confusion counts are retained in ANNEX D.2 (*Patient 4 Independent Confusion Counts by Hepatic View*). The separation prevents internal selection evidence from being confused with independent performance.

### 3.5 Computational Performance

Table XIII reports processing speed on the workstation specified in Methodology. The three segmentation networks were also measured individually, but the complete serial branch is the value that represents the operational *longitudinal-view* pipeline. The DINOv2 values include preprocessing, feature extraction, temporal summarization, classifier inference, and binary decision generation.

**TABLE XIII**  
Measured processing performance of the final branches

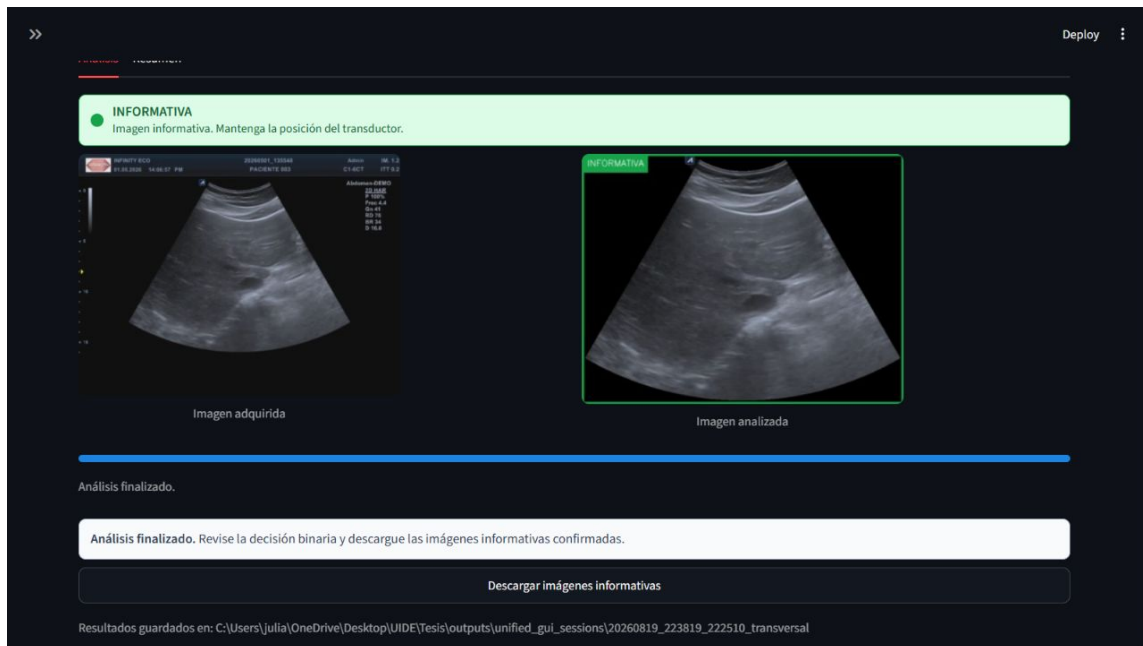
| Processing branch              |                          | Mean time           | Measured rate       | Interpretation                             |
|--------------------------------|--------------------------|---------------------|---------------------|--------------------------------------------|
| ROI segmentation               |                          | 9.80 ms/frame       | 102.1 FPS           | Individual model                           |
| Liver segmentation             |                          | 9.58 ms/frame       | 104.4 FPS           | Individual model                           |
| LA segmentation                |                          | 10.89 ms/frame      | 91.8 FPS            | Individual model                           |
| Complete pipeline              | <i>longitudinal-view</i> | 51.40 ms/frame      | 19.5 FPS            | Three models plus post-processing and rule |
| <i>Transverse-view</i> branch  | DINOv2                   | 154.3 ms/evaluation | 6.48 evaluations/s  | One of every five source frames evaluated  |
| <i>Oblique-view</i> branch     | DINOv2                   | 77.5 ms/evaluation  | 12.91 evaluations/s | One of every five source frames evaluated  |
| <i>Hepatorenal-view</i> branch | DINOv2                   | 61.7 ms/evaluation  | 16.21 evaluations/s | One of every five source frames evaluated  |

Note: These measurements describe interactive processing of recorded videos on the verified workstation; they do not demonstrate continuous live acquisition from an ultrasound device.

The complete *longitudinal-view* branch was slower than any individual segmentation network because ROI, liver, and LA inference were executed serially before post-processing and decision logic. The DINOv2 branches also operated below 30 evaluations/s and used a stride of five. Accordingly, the supported term is interactive recorded-video processing rather than validated real-time acquisition.

### 3.6 Graphical Interface

The GUI verified the integrated workflow. It processes recorded videos and updates two fixed panels with the acquired *frame* and the corresponding analyzed image. The user-facing decision is green for a confirmed informative capture, yellow for a potentially informative or uncertain image requiring a fine probe adjustment, and red for insufficient view evidence requiring repositioning; the initial DINOv2-Small warm-up is shown neutrally as analysis in progress. The analysis view does not display *frame* indices, LA area, probability, raw result, capture count, or speed. Once the video finishes, it presents an explicit completion notice and a control for downloading the confirmed informative images; the technical CSV and session summary remain stored for traceability. It does not acquire directly from an ultrasound machine or camera. Figure 18 documents the implemented states with real sequences: panel (a) is yellow in the *longitudinal view*, panel (b) is a confirmed green *transverse-view* capture, panel (c) is red for a non-informative *oblique-view* sequence, and panel (d) shows the completion notice and download control.



**Fig. 18.** Final graphical interface: (a) yellow fine-adjustment feedback with acquired and analyzed longitudinal images; (b) green confirmed transverse capture; (c) red feedback for insufficient oblique-view evidence; and (d) end-of-analysis notice and informative-image download control.

Source: Own implementation.

The verified interface functions and execution command are presented in ANNEX C.1 (*Graphical Interface Workflow, Stored Outputs, and Downloadable Captures*).

### 3.7 Discussion

The independent evaluation demonstrated preliminary technical feasibility but also revealed substantial differences among views. Globally, 206 of 322 binary outputs were classified correctly, with 37.3% sensitivity, 90.7% specificity, 61.2% macro-F1, and 64.0% *balanced accuracy*. High specificity combined with low sensitivity indicates conservative acceptance: the system rejected most non-informative outputs but failed to accept many *frames* that the radiologist considered informative. For acquisition assistance, this behavior could make a user continue searching after an acceptable image has already been obtained. It reduces false acceptance at the cost of longer acquisition and possible user uncertainty. The three-color interface communicates that uncertainty but does not change these binary performance values: yellow is an abstention and adjustment state, not a corrected prediction or a third radiologist label.

The segmentation results explain much of the *longitudinal-view* behavior. ROI and liver reached internal Dice scores of 0.9929 and 0.9416, whereas LA reached 0.8425 overall Dice and 0.8198 Dice in positive-mask cases. ROI and liver occupied larger and more continuous regions; LA was smaller, was not visible in every image, and was absent from 46 internal-test masks. Small localization errors therefore had a larger proportional effect on LA, while empty masks exposed false-positive behavior. Weak boundaries, speckle, shadowing, and small target structures are established difficulties in ultrasound segmentation [15, 16, 33, 34]. Because LA is mandatory in the final deterministic rule, an LA false negative propagates directly to the image result even when ROI and liver are detected. This accounts for the gap between strong internal ROI/liver overlap and the failure to accept the 101 independent informative longitudinal outputs. Mapping *liver\_without\_reference* to yellow is therefore preferable to a red rejection in the user interface: it communicates that the main hepatic field is already present while explicitly requesting a fine inclination adjustment to strengthen the LA-dependent evidence. This design reduces the severity of the feedback but does not conceal or repair the underlying LA failure.

The internal segmentation split was grouped by video, but the same development patients remained represented across partitions. Subject-specific anatomy, acquisition settings, and ultrasound appearance could therefore make the internal results optimistic. Patient 4 provides more relevant transfer evidence because it was excluded from training, model selection, and threshold derivation. However, Patient 4 had no pixel-level ROI, liver, or LA masks. The longitudinal failures can be associated operationally with the LA-dependent stage, but they can-

not be decomposed quantitatively into external segmentation error and rule error. The internal segmentation values should consequently be interpreted as component-selection evidence rather than patient-independent performance.

For the other three views, frozen DINOv2-Small representations allowed classical classifiers to be fitted without dense anatomical masks. This is consistent with the role of self-supervised representations when labeled medical datasets are limited [13, 37]. Performance nevertheless differed by view. The *transverse view* reached 100.0% recall but only 25.0% specificity, showing that Random Forest frequently accepted non-informative examples. No binary errors were observed for the *oblique view* or *hepatorenal view*; however, each result came from only one informative and one non-informative independent sequence. For these branches, yellow is justified by the calibrated interval  $0.35 < P(\text{informative}) < 0.65$  and by pending temporal confirmation: the available representation is not sufficiently decisive for red or green, so the interface asks for a small positional or angular adjustment while preserving the current anatomical window. Previous liver-view recognition studies have also reported promising results with learned image representations, but their anatomical targets, datasets, and validation schemes differ and their values cannot be used as direct benchmarks for this work [8, 12, 13].

The qualitative cases were consistent with the quantitative results. The representative longitudinal false negative contained ROI and liver predictions but did not satisfy the LA condition. The representative transverse false positive came from a non-informative sequence whose global representation was assigned to the informative class. The correct oblique and hepatorenal examples matched the appearances represented in their limited development data. These cases identify plausible failure locations, but they do not add independent evidence beyond the confusion counts.

Computational performance must be interpreted at complete-pipeline level. The individual longitudinal networks reached approximately 91.8–104.4 FPS, but the serial three-network pipeline with post-processing reached 19.5 FPS. The DINOv2 branches reached 6.48, 12.91, and 16.21 evaluations/s for the *transverse view*, *oblique view*, and *hepatorenal view*, respectively, while evaluating one of every five source *frames*. Consistency of the final decision functions between offline analysis and the GUI was verified using shared code and configuration; the interface does not modify the algorithmic result. These measurements support interactive processing of recorded videos, not continuous live acquisition.

The results support the preliminary feasibility of identifying protocol-defined informative

hepatic-ultrasound images acquired after short procedural training. They do not demonstrate diagnostic validity, replacement of a specialist, or generalization to other operators, devices, or clinical environments. Tele-ultrasound and guided acquisition studies show why such assistance could eventually be useful, but they also emphasize the need for structured protocols, specialist oversight, and validation in the intended setting [1, 2]. Tele-ultrasound, supervised home acquisition, and frequent monitoring therefore remain possible future applications rather than demonstrated outcomes of this prototype.

The principal limitations are three development patients and one independent-evaluation patient, video-grouped but not patient-independent internal segmentation partitions, only three independent sequences per view, absence of independent pixel masks for the longitudinal branch, thresholds derived from a small development dataset, and binary *ground truth* established by one expert radiologist. The original acquisition-quality folders were development metadata rather than independent clinical labels. In addition, the study used one acquisition environment and one ultrasound equipment configuration, and the GUI processed recorded videos without a direct device stream. These limitations restrict the conclusions to technical feasibility under the defined dataset and protocol.

## Chapter IV

### Conclusions and Recommendations

#### 4.1 Conclusions

The first objective was to define the view-specific informativity criteria and implement their software analysis workflow. This objective was achieved through a four-view workflow that associates recorded videos with the *longitudinal view*, *transverse view*, *oblique view*, and *hepatorenal view*, and routes each view to its corresponding computational branch. The result is an informative-image identification prototype; because transducer position and protocol compliance were not directly sensed or clinically validated, it does not demonstrate standardized acquisition.

The second objective was to implement a graphical user interface that provides feedback during recorded-video processing. This objective was achieved through an interface that accepts ultrasound videos by upload or local path, allows view selection, processes *frames* sequentially, and updates two fixed panels with the acquired and analyzed images. Its classifiable feedback uses green for a confirmed capture, yellow for a potentially informative or uncertain image requiring fine adjustment, and red for insufficient view evidence requiring repositioning; the status remains neutral only while the DINOv2-Small window is initialized. It displays a completion notice and permits downloading confirmed informative images, while the technical CSV and session evidence are stored without exposing area, probability, raw-state, capture-count, or speed statistics in the analysis panel. The three colors are operational guidance states and do not replace the binary expert label. The interface does not acquire live ultrasound data directly, but it verifies the interaction flow for processing recorded sequences.

The third objective was to develop deep-learning-based algorithms for the identification of hepatic ultrasound images that satisfy the implemented acquisition criteria. This objective was achieved using two complementary strategies: segmentation and deterministic verification for the *longitudinal view*, and frozen DINOv2-Small *embeddings* with view-specific binary classifiers for the *transverse view*, *oblique view*, and *hepatorenal view*. The final longitudinal component identified ROI, liver, and an anechoic lumen without assigning LA to a uniquely identified vessel. The DINOv2-Small component used Random Forest, logistic regression, and distance-weighted  $k$ -NN, respectively. The final binary output indicates conformity with the

radiologist-defined informativity label within this dataset and protocol, not diagnosis or guaranteed diagnostic value.

The fourth objective was to evaluate the consistency and feasibility of the proposed system. It was addressed within the available software scope through controlled segmentation experiments, ablation studies, transfer-learning comparisons, leave-one-patient-out DINOv2 evaluation, inference benchmarks, GUI sessions, and the independent Patient 4 evaluation. The independent results were 37.3% sensitivity and 90.7% specificity, with macro-F1 values of 33.3%, 56.4%, 100.0%, and 100.0% for the *longitudinal view*, *transverse view*, *oblique view*, and *hepatorenal view*, respectively. These tests demonstrate functional feasibility but do not establish prospective, multicenter, or clinical validation. Only four patients were valid, the internal segmentation split shared patients between partitions, Patient 4 lacked longitudinal pixel-level masks, and each DINOv2 view had only one informative and one non-informative independent sequence. LA detection remained the longitudinal bottleneck, transverse false positives were observed, and the serial three-model component still requires latency optimization.

## 4.2 Recommendations

Future work should annotate Patient 4 and additional patients with ROI, liver, and LA masks before recalibrating the longitudinal thresholds. This would allow external Dice, IoU, recall, and false-positive metrics to be reported without using the independent evaluation set as an informal tuning set.

A future version should include a dedicated monitoring software module that automatically generates a formal timing report. This report should include model loading time, processing time per *frame*, post-processing time, FPS, session duration, and exported evidence paths.

The longitudinal branch should be optimized before live acquisition is attempted. Possible improvements include a unified multi-output segmentation model, conditional execution of LA only after ROI and liver are verified, model compression, asynchronous inference, or lower-frequency analysis with temporal interpolation.

The DINOv2 component should be evaluated with more videos per view, more operators, additional equipment, and more patients. The 0.50 binary decision boundary should be calibrated on a larger validation set and reported with confidence intervals or uncertainty measures. The *five-embedding* representation should be evaluated against *single-frame* and alternative aggregation strategies because it defines the temporal context presented to the



classifier.

The longitudinal rule should also be recalibrated across equipment and acquisition settings. The area, standard-deviation, entropy, and border thresholds were derived from development distributions dominated by a small number of patients, while the  $15 \times 15$  and  $9 \times 9$  kernels were heuristic neighborhoods. Future evaluation should quantify how these parameters change sensitivity, false rejection, and false capture before any clinical deployment claim is made.

## **ANNEX A: Acquisition Protocol, Acquired Videos, and Prepared Dataset**

### **A.1 Hepatic-Ultrasound Acquisition Guide for the Four Views**

The acquisition guide defined four targets. The *longitudinal view* used a right-midclavicular longitudinal orientation and sought hepatic parenchyma and anechoic vascular references. The *transverse view* used a horizontal right-subcostal orientation and sought broad hepatic coverage with vascular branches. The *oblique view* used a cranial and medial subcostal orientation and sought deep parenchyma with diagonally oriented vascular structures. The *hepatorenal view* used an intercostal right-flank orientation and required simultaneous visualization of the liver, right kidney, and their interface. These four definitions are the acquisition targets used for labeling and software evaluation in the thesis.

### **A.2 Repository of Raw Ultrasound Videos Used in Development and Independent Evaluation**

The expected evidence location for this annex is the repository directory `data/01_acquired_videos/`, which separates the twelve Patient 4 sequences from the development-video folders. Individual ultrasound videos are not publicly distributed because they are study data subject to institutional authorization. The public repository records this restriction and the required organization at <https://github.com/Julian7073/hepatic-ultrasound-acquisition-assistance/blob/v1.1.0-thesis/data/README.md>. Table XIV defines the exact content and organization used in the controlled research environment.

**TABLE XIV**  
Repository folders for acquisition and prepared data

| Repository folder         | Evidence level         | Required content                                                              |
|---------------------------|------------------------|-------------------------------------------------------------------------------|
| data/01_acquired_videos/  | Raw acquisition        | Patient code, view, video identifier, and original recorded sequence.         |
| data/02_extracted_frames/ | Decoded source data    | Ordered $1024 \times 768$ frames with temporal index.                         |
| data/03_annotations/      | Prepared labels        | COCO-compatible ROI, liver, and LA annotations and radiologist binary labels. |
| data/04_splits/           | Experimental partition | Development, validation, internal-test, and independent-evaluation manifests. |

Note: Patient identifiers in filenames may retain the audited repository codes, while the thesis text uses Patients 1–4.

### A.3 Repository of Extracted Frames, Annotations, and Dataset Splits

The controlled evidence locations for this annex are the following repository directories:

- data/02\_extracted\_frames/ for decoded source frames;
- data/03\_annotations/ for segmentation and radiologist labels; and
- data/04\_splits/ for patient-level experimental partitions.

These patient-level files are not published; the access restriction and expected directory structure are documented at <https://github.com/Julian7073/hepatic-ultrasound-acquisition-assistance/blob/v1.1.0-thesis/data/README.md>. The public aggregate tables permit audit of the reported numerical results without exposing individual images or annotations.

## ANNEX B: Reproducible Implementation of the Longitudinal and DINOv2-Small Analysis Pipelines

### B.1 Source Code for *Longitudinal-View* Segmentation and Deterministic Decision

This annex corresponds specifically to segmentation and deterministic processing of the *longitudinal view*. The repository separates model development and internal testing from independent Patient 4 validation under 03\_CODIGO\_ENTRENAMIENTO\_Y\_TEST/01\_DESARROLLO/01\_VISTA\_LONGITUDINAL/ and 03\_CODIGO\_ENTRENAMIENTO\_Y\_TEST/02\_VALIDACION\_INDEPENDIENTE/01\_VISTA\_LONGITUDINAL/. These directories contain training, threshold derivation, checkpoint loading, preprocessing, ROI/liver/LA inference, mask post-processing, GLCM and border functions, and independent evidence generation. The frozen organized code is available at [https://github.com/Julian7073/hepatic-ultrasound-acquisition-assistance/tree/v1.1.0-thesis/03\\_CODIGO\\_ENTRENAMIENTO\\_Y\\_TEST/01\\_DESARROLLO/01\\_VISTA\\_LONGITUDINAL](https://github.com/Julian7073/hepatic-ultrasound-acquisition-assistance/tree/v1.1.0-thesis/03_CODIGO_ENTRENAMIENTO_Y_TEST/01_DESARROLLO/01_VISTA_LONGITUDINAL) and [https://github.com/Julian7073/hepatic-ultrasound-acquisition-assistance/tree/v1.1.0-thesis/03\\_CODIGO\\_ENTRENAMIENTO\\_Y\\_TEST/02\\_VALIDACION\\_INDEPENDIENTE/01\\_VISTA\\_LONGITUDINAL](https://github.com/Julian7073/hepatic-ultrasound-acquisition-assistance/tree/v1.1.0-thesis/03_CODIGO_ENTRENAMIENTO_Y_TEST/02_VALIDACION_INDEPENDIENTE/01_VISTA_LONGITUDINAL).

The following pseudocode links the model outputs to the binary evaluation and three-color interface states without reproducing line-by-line source code:

for each evaluated frame:

```

predict and post-process ROI, liver, and LA masks
if ROI is absent:
    raw_state = no_structure                -> red
else if liver is absent or liver/ROI < 0.15:
    raw_state = partial_liver              -> red
else:
    compute LA area and standard deviation
    compute GLCM entropy on ROI AND dilated_LA
    compute border evidence on the LA ring
    if every final rule condition is satisfied:
        raw_state = capture
    else:
```

```

        raw_state = liver_without_reference -> yellow
    stabilize the last five raw decisions
    if capture occurs three consecutive times:
        visible_state = green and save subject to cooldown
    else if the stabilized state has adequate liver but not LA evidence:
        visible_state = yellow
    else:
        visible_state = red
    binary_evaluation = informative only for capture; otherwise non-informative

```

## B.2 Source Code for DINOv2-Small Feature Extraction and View-Specific Classification

This annex corresponds specifically to the *transverse view*, *oblique view*, and *hepatorenal view*. The repository separates development and internal testing under 03\_CODIGO\_ENTRENAMIENTO\_Y\_TEST/01\_DESARROLLO/02\_OTRAS\_VISTAS/ from independent Patient 4 validation under 03\_CODIGO\_ENTRENAMIENTO\_Y\_TEST/02\_VALIDACION\_INDEPENDIENTE/02\_OTRAS\_VISTAS/; each stage is then divided by view. These directories contain deterministic fan cropping, DINOv2-Small feature extraction, five-*embedding* aggregation, Random Forest/logistic-regression/7-NN fitting, leave-one-patient-out comparison, probability thresholds, temporal confirmation, and independent evaluation. The frozen organized implementation is available at [https://github.com/Julian7073/hepatic-ultrasound-acquisition-assistance/tree/v1.1.0-thesis/03\\_CODIGO\\_ENTRENAMIENTO\\_Y\\_TEST](https://github.com/Julian7073/hepatic-ultrasound-acquisition-assistance/tree/v1.1.0-thesis/03_CODIGO_ENTRENAMIENTO_Y_TEST).

The DINOv2-Small operational path is summarized below. The 0.50 binary boundary is intentionally kept separate from the action thresholds:

```

for each evaluated frame:
    preprocess the fan crop or full frame
    z = frozen_DINOv2_Small_CLS(frame)                # 384 values
    append z to the five-embedding buffer
    if fewer than five embeddings are available:
        visible_state = neutral warm-up
    else:
        phi = concatenate(mean(buffer), population_std(buffer))
        p = view_specific_classifier.predict_proba(phi)

```

```

binary_evaluation = informative if p >= 0.50 else non-informative
if p <= 0.35:
    visible_state = red                # adjust
else if p < 0.65:
    visible_state = yellow             # doubtful
else if fewer than three consecutive capture actions:
    visible_state = yellow             # pending confirmation
else:
    visible_state = green              # confirmed capture
    save subject to the ten-evaluation cooldown

```

### B.3 Reproduction Commands and Parameter-to-Code Traceability

Table XV lists the principal commands. They are executed from the repository root after installing the declared dependencies. The complete tagged repository is available at <https://github.com/Julian7073/hepatic-ultrasound-acquisition-assistance/tree/v1.1.0-thesis>; the three segmentation checkpoints and their SHA-256 digests are available at <https://github.com/Julian7073/hepatic-ultrasound-acquisition-assistance/releases/tag/v1.1.0-thesis>.

**TABLE XV**  
Main reproduction commands

| Task                              | Command                                                                                                                                                        |
|-----------------------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Run GUI                           | <code>python -m streamlit run Codigos_Pipeline_Experimental_Segmentacion/gui_adquisicion_hepatica.py</code>                                                    |
| Patient 4 longitudinal evaluation | <code>python Codigos_Pipeline_Experimental_Segmentacion/evaluate_p005_longitudinal_final.py</code><br><code>--frame_stride 1 --save_overlays --save_csv</code> |
| Freeze evidence package           | <code>python Codigos_Pipeline_Experimental_Segmentacion/scripts/12_freeze_final_project.py</code>                                                              |

Note: Commands should be adapted only to the local repository path; model and decision parameters remain frozen.

Table XVI maps the principal decision parameters to their source functions and output evidence.

**TABLE XVI**  
Code and output traceability for principal decision parameters

| Parameter / method                                             | Value                                                                                                             | File                                     | Function / class                                    | Approx. line | Output evidence                               |
|----------------------------------------------------------------|-------------------------------------------------------------------------------------------------------------------|------------------------------------------|-----------------------------------------------------|--------------|-----------------------------------------------|
| LA area, intensity standard deviation, and GLCM entropy limits | 874 px; 43.330219; 6.125722                                                                                       | 05_definir_umbral_ac_eptabilidad.py      | main                                                | 28–75        | threshold_summary.csv                         |
| GLCM region and descriptors                                    | $ROI \cap LA_{dilated15}$ ; 32 levels; four offsets; contrast, entropy, homogeneity, energy                       | longitudinal_quality_rules.py            | compute_valid_lumen_region; compute_glcm_features   | 70–174       | glcm_longitudinal_metrics_with_thresholds.csv |
| Border ring and evidence branches                              | $9 \times 9$ ring; 3.0, 0.0375, 6.324555, 35.0; logical OR                                                        | 11_ajustar_regla_ac_eptabilidad_borde.py | compute_border_metrics; derive_thresholds           | 57–164       | border_rule_threshold_summary.csv             |
| Frozen longitudinal rule and states                            | Area, standard deviation, entropy, border; four raw states                                                        | longitudinal_quality_rules.py            | evaluate_lumen_rule; finalize_longitudinal_decision | 304–379      | longitudinal_decision_config.json             |
| DINOv2 classifier input and warm-up                            | Five 384-feature <i>embeddings</i> ; mean + population standard deviation; 768 features; four warm-up evaluations | binary_inference.py                      | BinaryDinoVideoPredictor.predict_rgb                | 72–104       | 09_binary_model_manifest.csv                  |

| Parameter / method                            | Value                                                                                                                                                                     | File                        | Function / class                                                       | Approx. line | Output evidence                                |
|-----------------------------------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-----------------------------|------------------------------------------------------------------------|--------------|------------------------------------------------|
| View-specific classifier hyperparameters      | classi- RF: 300 trees, leaf 2; LR: $C = 1$ , 5000 iterations; 7-NN: distance weighting                                                                                    | classification.py           | create_classifier                                                      | 23–60        | 08_binary_winners_by_view.csv                  |
| Operational thresholds                        | action $\text{red/adjust} \leq 0.35$ ; $\text{yellow/doubtful} (0.35, 0.65)$ ; $\text{green/capture} \geq 0.65$ after confirmation                                        | binary_temporal.py          | calibrate_abstention_thresholds                                        | 149–212      | 09_binary_abstention_thresholds.csv            |
| Independent decision                          | binary $P(\text{informative}) \geq 0.50$                                                                                                                                  | binary_temporal.py          | clear_probabilities                                                    | 62–80        | 09_binary_p005_metrics.csv                     |
| GUI state mapping, stabilization, and storage | Longitudinal: red for absent/partial anatomy, yellow for inadequate LA evidence, green for confirmed capture; DINOv2-Small: thresholds above; confirmation 3; interval 10 | gui_adquisicion_hepatica.py | user_state_from_decision; guidance_message; run_longitudinal; run_dino | 115–1058     | GUI screenshot, session CSV, and capture files |

Note: RF and LR denote Random Forest and logistic regression. File and evidence names are relative to the repository folders identified in Sections B.1 and B.2. The 0.35/0.65 limits control the visible operational feedback; independent binary metrics use 0.50.



## **ANNEX C: Graphical Interface Operation and Integrated Verification**

### **C.1 Graphical Interface Workflow, Stored Outputs, and Downloadable Captures**

The graphical interface is implemented directly in `Codigos_Pipeline_Experimental_Segmentacion/gui_adquisicion_hepatica.py`. It accepts a local recorded video, requires selection of one of four views, updates the acquired and analyzed images in place, uses green, yellow, or red acquisition feedback, displays an end-of-analysis notice, and offers a ZIP download when confirmed informative images exist. Green confirms capture, yellow requests a fine adjustment for a potentially informative or uncertain image, and red requests repositioning when the view evidence is insufficient. The initialization stage remains neutral and is not a fourth class. Session CSV files and summaries retain the technical evidence. The frozen interface source is available at [https://github.com/Julian7073/hepatic-ultrasound-acquisition-assistance/blob/v1.1.0-thesis/Codigos\\_Pipeline\\_Experimental\\_Segmentacion/gui\\_adquisicion\\_hepatica.py](https://github.com/Julian7073/hepatic-ultrasound-acquisition-assistance/blob/v1.1.0-thesis/Codigos_Pipeline_Experimental_Segmentacion/gui_adquisicion_hepatica.py).

### **C.2 End-to-End Verification Protocol for the Integrated System**

Table XVII lists the ten properties checked in the final integrated implementation and the evidence used to verify each one.

**TABLE XVII**  
Verification protocol for the final integrated implementation

| Test | Property checked                             | Evidence                                                     | Result |
|------|----------------------------------------------|--------------------------------------------------------------|--------|
| T1   | Load a recorded ultrasound video.            | Decoded <i>frames</i> and ordered indices.                   | Passed |
| T2   | Select one of four hepatic views.            | Four routed analysis branches.                               | Passed |
| T3   | Load final model and classifier files.       | Three checkpoints and three classifier bundles.              | Passed |
| T4   | Segment ROI, liver, and LA.                  | Three binary masks.                                          | Passed |
| T5   | Use the shared GLCM region.                  | $ROI \cap LA_{dilated15}$ .                                  | Passed |
| T6   | Apply the frozen longitudinal rule.          | Configuration and rule output.                               | Passed |
| T7   | Produce three-state visible guidance.        | Green capture, yellow fine adjustment, or red repositioning. | Passed |
| T8   | Execute each DINOv2-Small classifier branch. | RF, LR, and 7-NN outputs.                                    | Passed |
| T9   | Save CSV and session summary.                | Reconciled stored records.                                   | Passed |
| T10  | Report processing measurements.              | Time and rate fields.                                        | Passed |

## ANNEX D: Internal-Selection and Independent-Evaluation Evidence

### D.1 Development-Patient Outputs for Model Selection

Table XVIII reports internal outputs by held-out development patient. They are selection evidence and must not be interpreted as independent Patient 4 performance.

**TABLE XVIII**  
Internal binary outputs represented in sampled *frames* by development patient

| Patient   | View                    | Videos | Frames | Informative / total | Non-informative / total |
|-----------|-------------------------|--------|--------|---------------------|-------------------------|
| Patient 1 | <i>Transverse view</i>  | 3      | 60     | 40/60 (66.7%)       | 20/60 (33.3%)           |
| Patient 1 | <i>Oblique view</i>     | 3      | 60     | 40/60 (66.7%)       | 20/60 (33.3%)           |
| Patient 1 | <i>Hepatorenal view</i> | 5      | 100    | 60/100 (60.0%)      | 40/100 (40.0%)          |
| Patient 2 | <i>Transverse view</i>  | 3      | 60     | 20/60 (33.3%)       | 40/60 (66.7%)           |
| Patient 2 | <i>Oblique view</i>     | 4      | 80     | 40/80 (50.0%)       | 40/80 (50.0%)           |
| Patient 2 | <i>Hepatorenal view</i> | 3      | 60     | 20/60 (33.3%)       | 40/60 (66.7%)           |
| Patient 3 | <i>Transverse view</i>  | 3      | 60     | 40/60 (66.7%)       | 20/60 (33.3%)           |
| Patient 3 | <i>Oblique view</i>     | 3      | 60     | 20/60 (33.3%)       | 40/60 (66.7%)           |
| Patient 3 | <i>Hepatorenal view</i> | 4      | 80     | 60/80 (75.0%)       | 20/80 (25.0%)           |

Note: Counts were reconstructed from the frozen selected-configuration leave-one-patient-out probabilities at the 0.50 binary boundary.

### D.2 Patient 4 Independent Confusion Counts by Hepatic View

Table XIX expands the central independent-performance table into TP, TN, FP, and FN counts for each view.

**TABLE XIX**  
Independent Patient 4 confusion counts by hepatic view

| <b>View</b>              | <b>TP</b> | <b>TN</b>  | <b>FP</b> | <b>FN</b>  | <b>Correct / total</b>           |
|--------------------------|-----------|------------|-----------|------------|----------------------------------|
| <i>Longitudinal view</i> | 0         | 101        | 0         | 101        | 101/202<br>(50.0%)               |
| <i>Transverse view</i>   | 20        | 5          | 15        | 0          | 25/40<br>(62.5%)                 |
| <i>Oblique view</i>      | 20        | 20         | 0         | 0          | 40/40<br>(100.0%)                |
| <i>Hepatorenal view</i>  | 20        | 20         | 0         | 0          | 40/40<br>(100.0%)                |
| <b>Total</b>             | <b>60</b> | <b>146</b> | <b>15</b> | <b>101</b> | <b>206/322</b><br><b>(64.0%)</b> |

Note: TP, TN, FP, and FN refer to the informative class and the expert-radiologist binary *ground truth*.

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